Nebico Oil Retention Project (3.3)

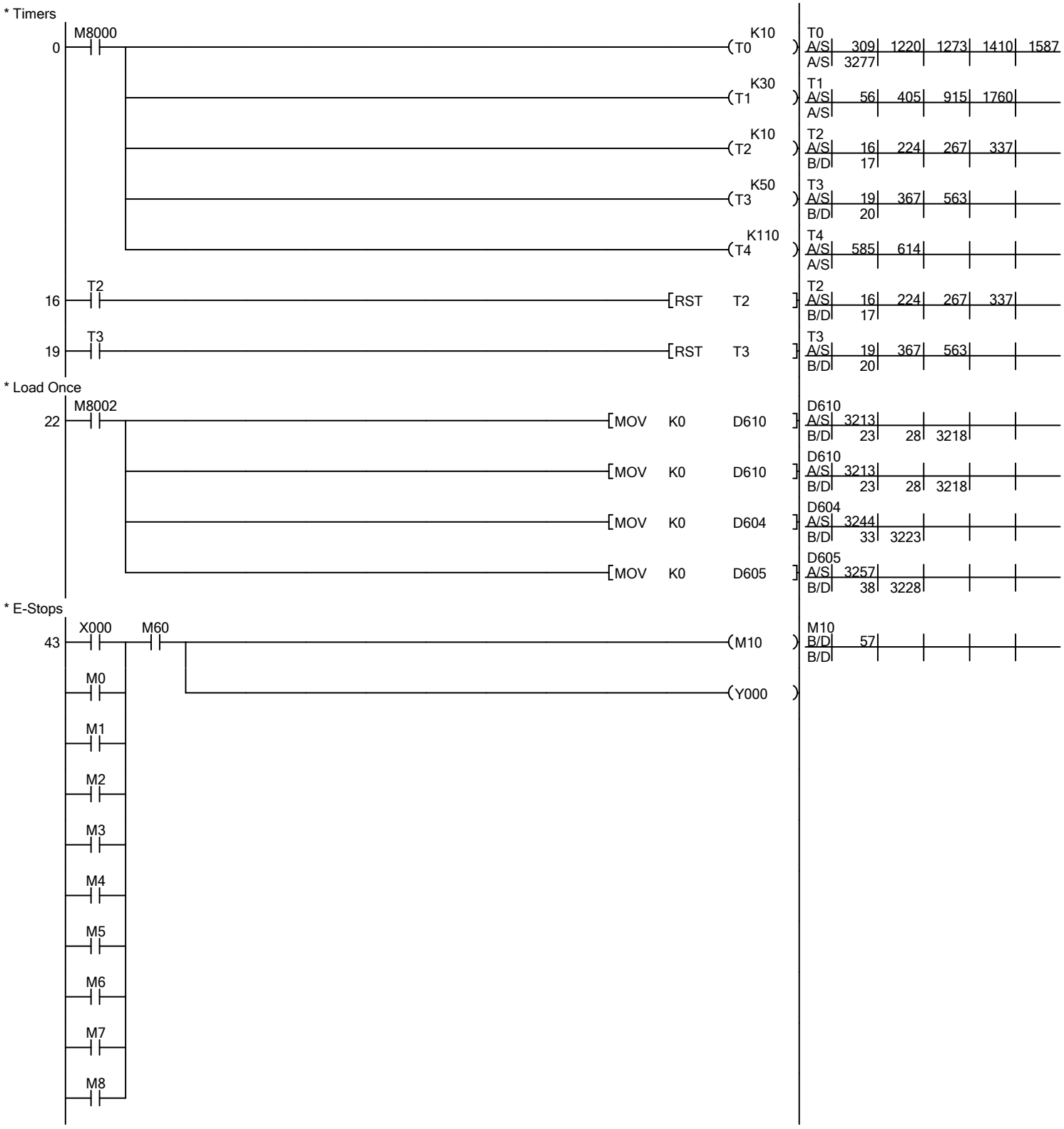
Coolmay L10S Ladder program
TK8050 x 3, TK8100 x 1

Modifications:

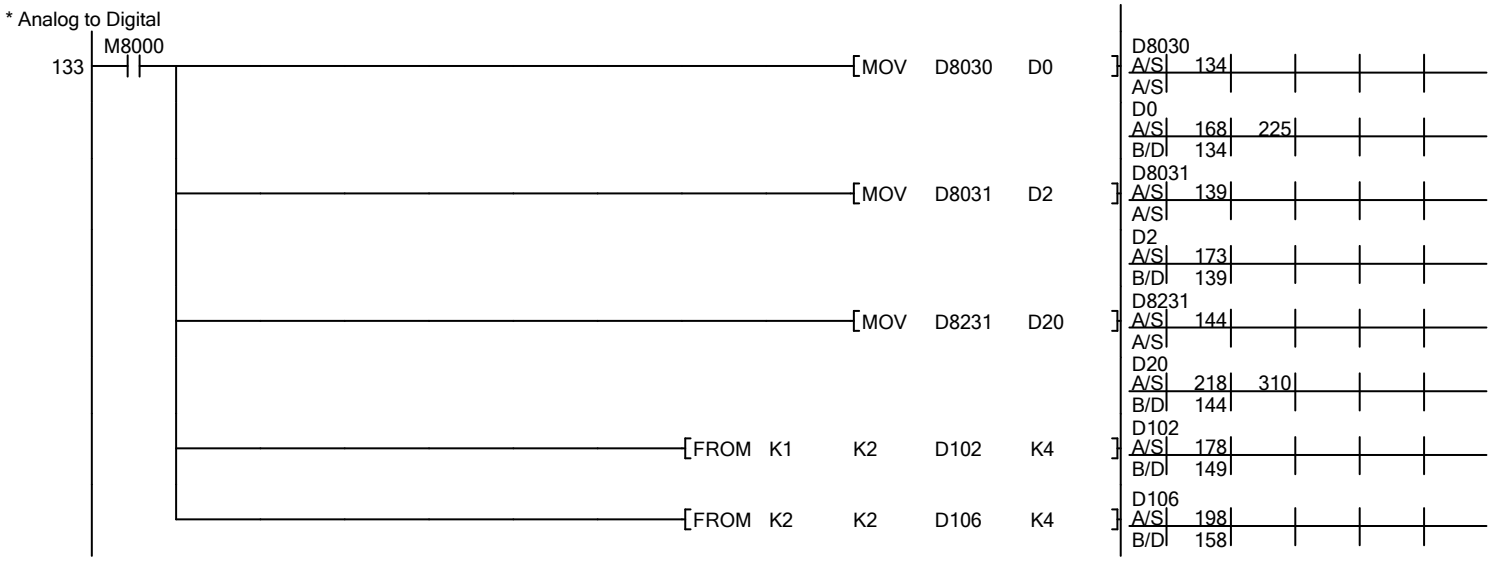
1. Pointer Functions to shorten bulk calculation
2. Revised Core logic (Mode Scheduler)
3. Revised Oil Distribution
4. Clean Segmentation

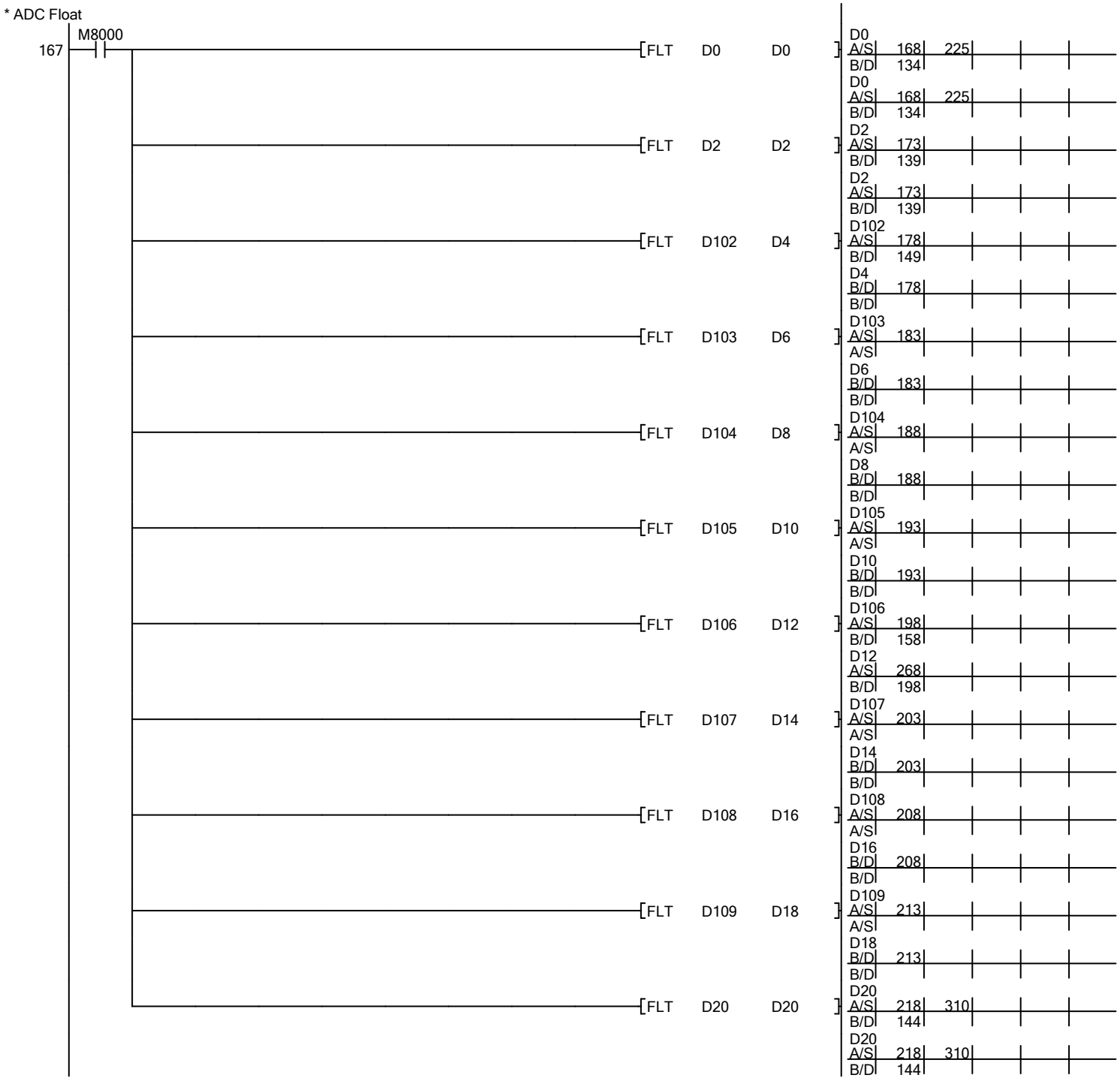
Prepared By: Sambridda Ranabhat (Architect & Programmer)

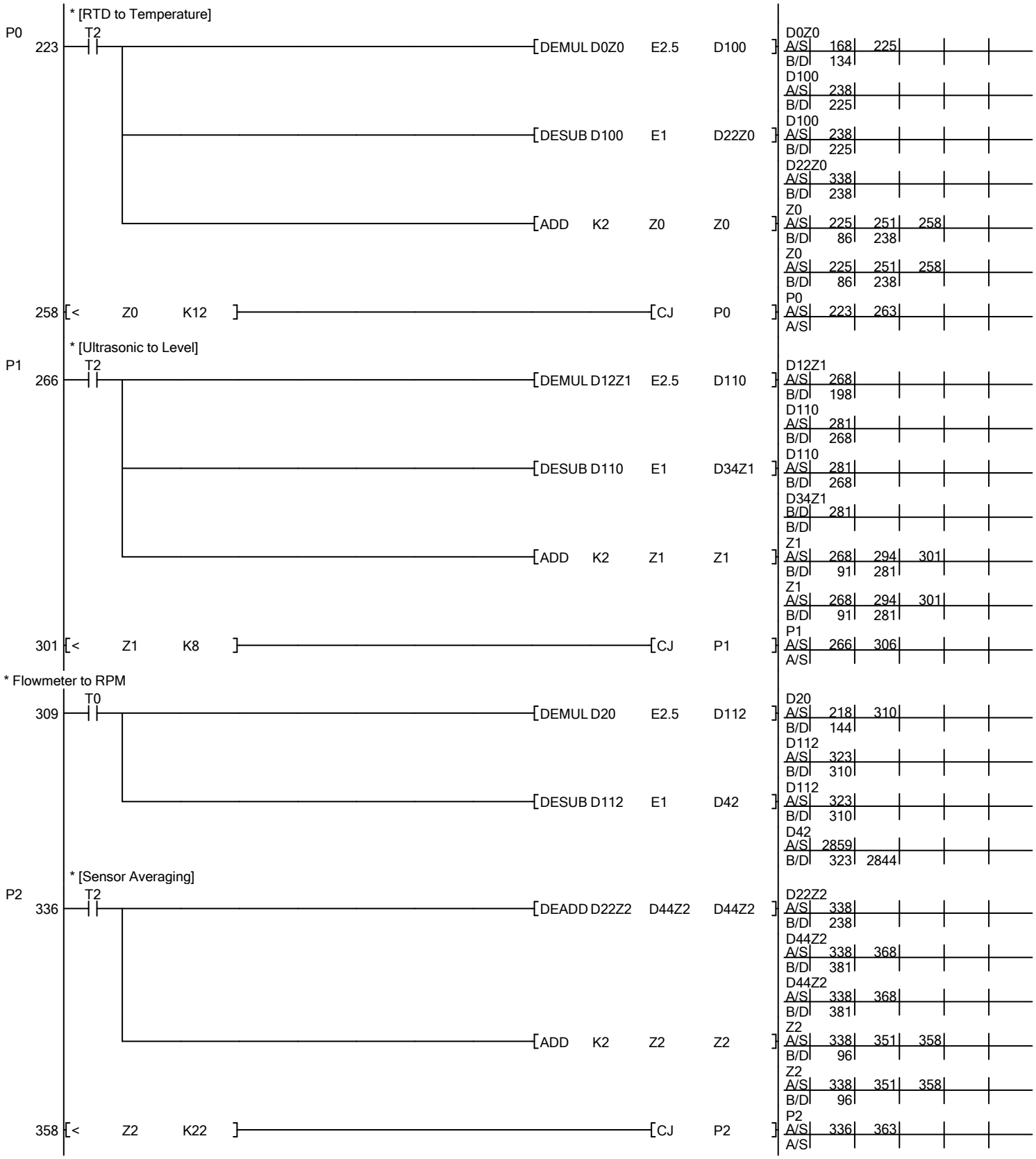
09/09/2026
Mechatronics Engineering .pvt .ltd

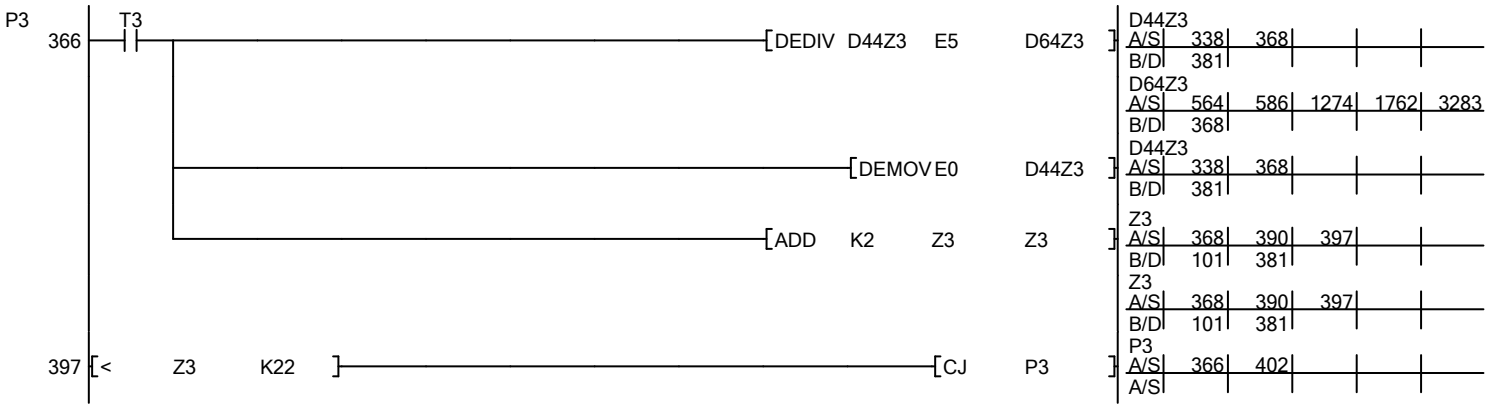

* Timers
0
M8000
K10
T0
T0
A/S 309 1220 1273 1410 1587
A/S 3277
K30
T1
T1
A/S 56 405 915 1760
A/S
K10
T2
T2
A/S 16 224 267 337
B/D 17
K50
T3
T3
A/S 19 367 563
B/D 20
K110
T4
T4
A/S 585 614
A/S
16
T2
RST T2
T2
A/S 16 224 267 337
B/D 17
19
T3
RST T3
T3
A/S 19 367 563
B/D 20
* Load Once
22
M8002
MOV K0 D610
D610
A/S 3213
B/D 23 28 3218
MOV K0 D610
D610
A/S 3213
B/D 23 28 3218
MOV K0 D604
D604
A/S 3244
B/D 33 3223
MOV K0 D605
D605
A/S 3257
B/D 38 3228
* E-Stops
43
X000
M60
M10
M10
B/D 57
B/D
Y000
M0
M1
M2
M3
M4
M5
M6
M7
M8

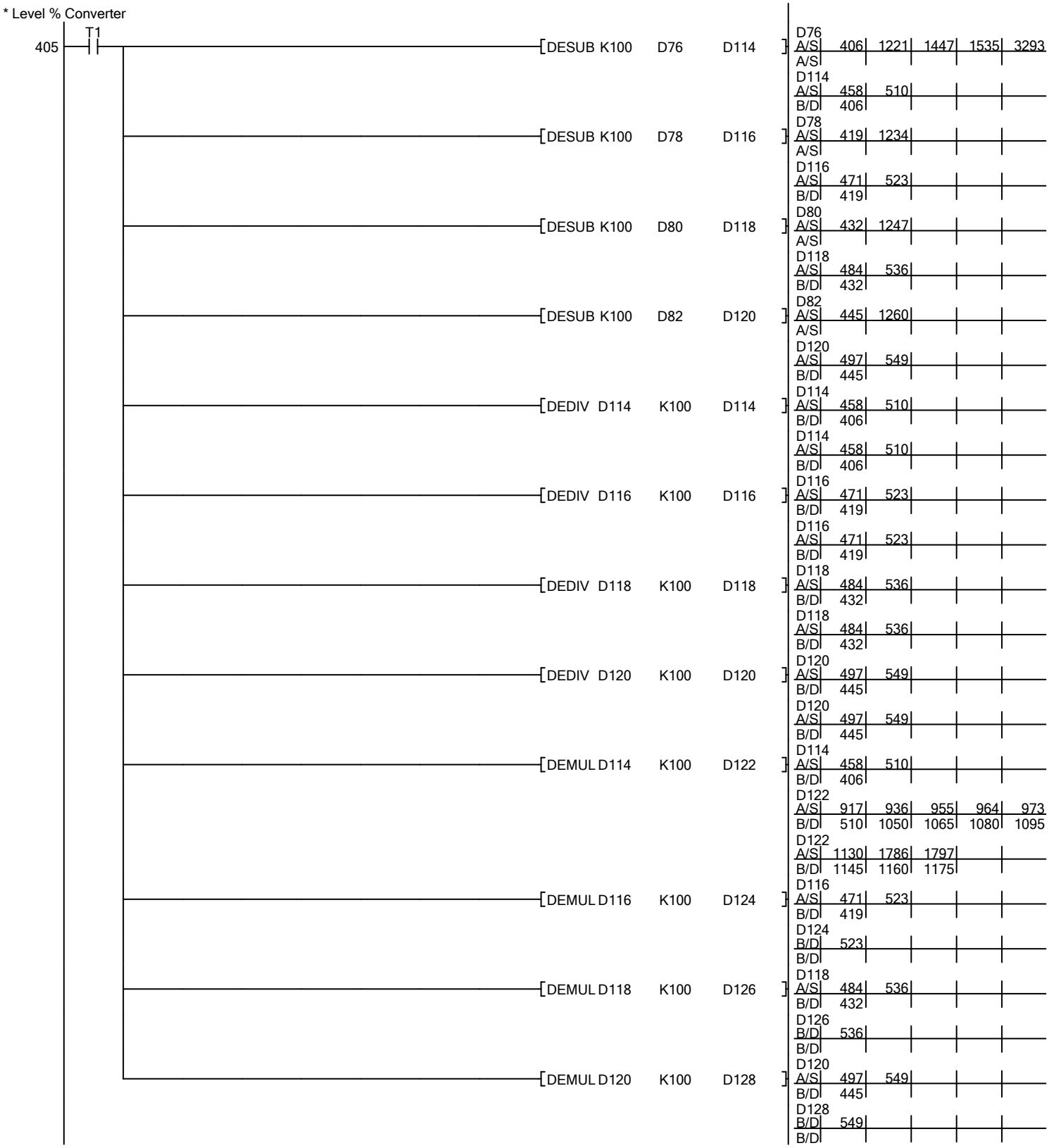

* Automation
56
T1
M10
X001
M11
Y001
X001
M12
M11
A/S 1038 1829 1848 1864 2365
A/S 2539 2544 2549 2554 2559
M11
A/S 2563 2567 2571 2575 2579
A/S 2582 2611 2959 3015 3020
M12
A/S 1210 2607 3012 3016 3050
A/S 3078 3081 3113 3116 3138
M12
A/S 3141
A/S
* Declarations
65
M8000
MOV K15 D8114
MOV K0 D550
MOV K0 D304
MOV K0 D303
MOV K0 Z0
MOV K0 Z1
MOV K0 Z2
MOV K0 Z3
MOV K0 Z4
MOV K0 Z5
MOV K97 D8400
MOV H1 D8401
SET M8192
MOV K1 D8397
D8114
B/D 66
B/D
D550
A/S 2745 2778 2811
B/D 71
D304
A/S 1561 2763 2796 2829
B/D 76 2772 2805 2838
D303
A/S 2653 2691 2729
B/D 81 2661 2699 2737
Z0
A/S 225 251 258
B/D 86 238
Z1
A/S 268 294 301
B/D 91 281
Z2
A/S 338 351 358
B/D 96
Z3
A/S 368 390 397
B/D 101 381
Z4
A/S 564 569 576
B/D 106
Z5
A/S 586 599 606
B/D 111
D8400
B/D 116
B/D
D8401
B/D 121
B/D
M8192
A/S 126
A/S
D8397
B/D 128
B/D

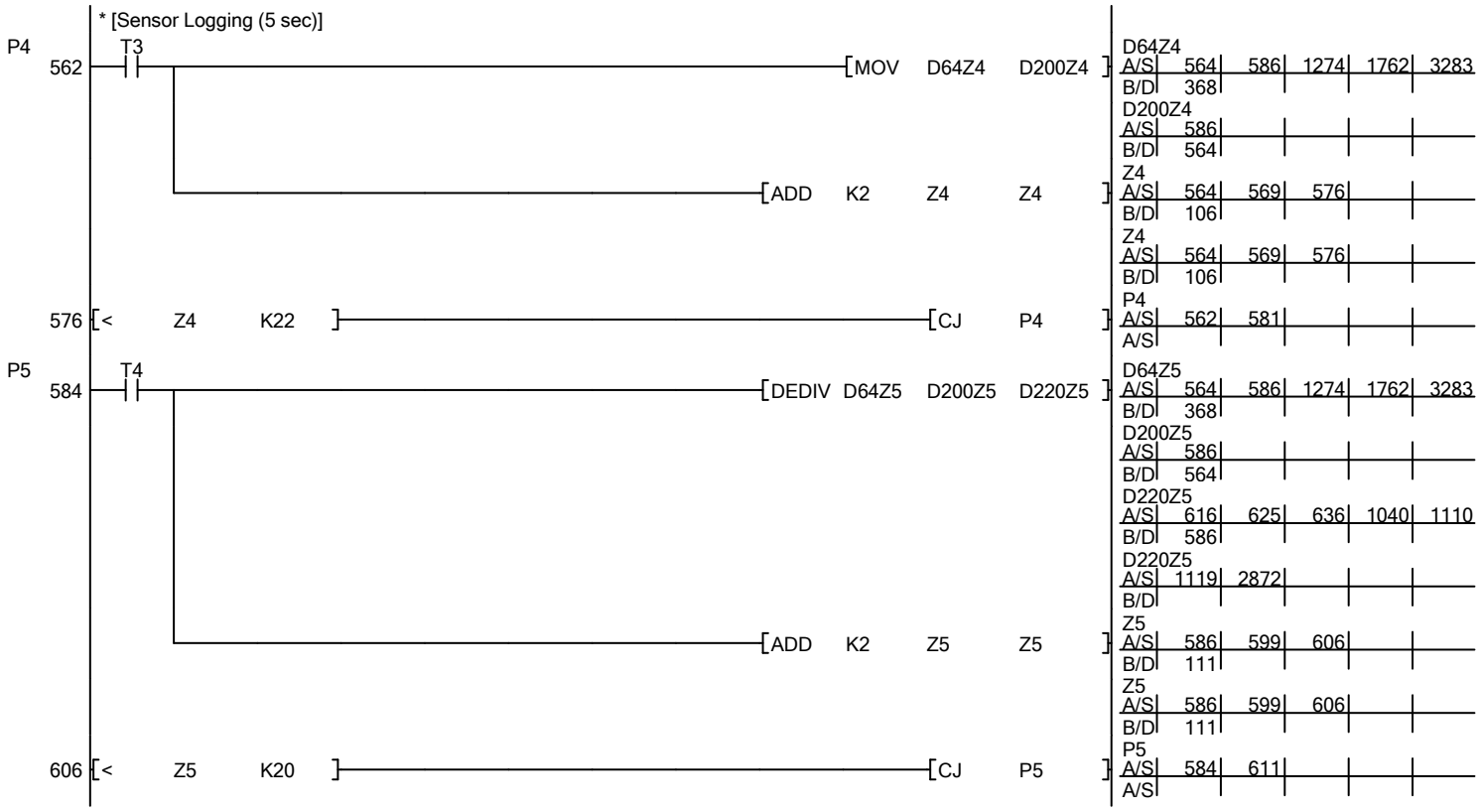

* Analog to Digital
M8000
133
MOV D8030 D0
MOV D8031 D2
MOV D8231 D20
FROM K1 K2 D102 K4
FROM K2 K2 D106 K4
D8030
A/S 134
A/S
D0
A/S 168 225
B/D 134
D8031
A/S 139
A/S
D2
A/S 173
B/D 139
D8231
A/S 144
A/S
D20
A/S 218 310
B/D 144
D102
A/S 178
B/D 149
D106
A/S 198
B/D 158

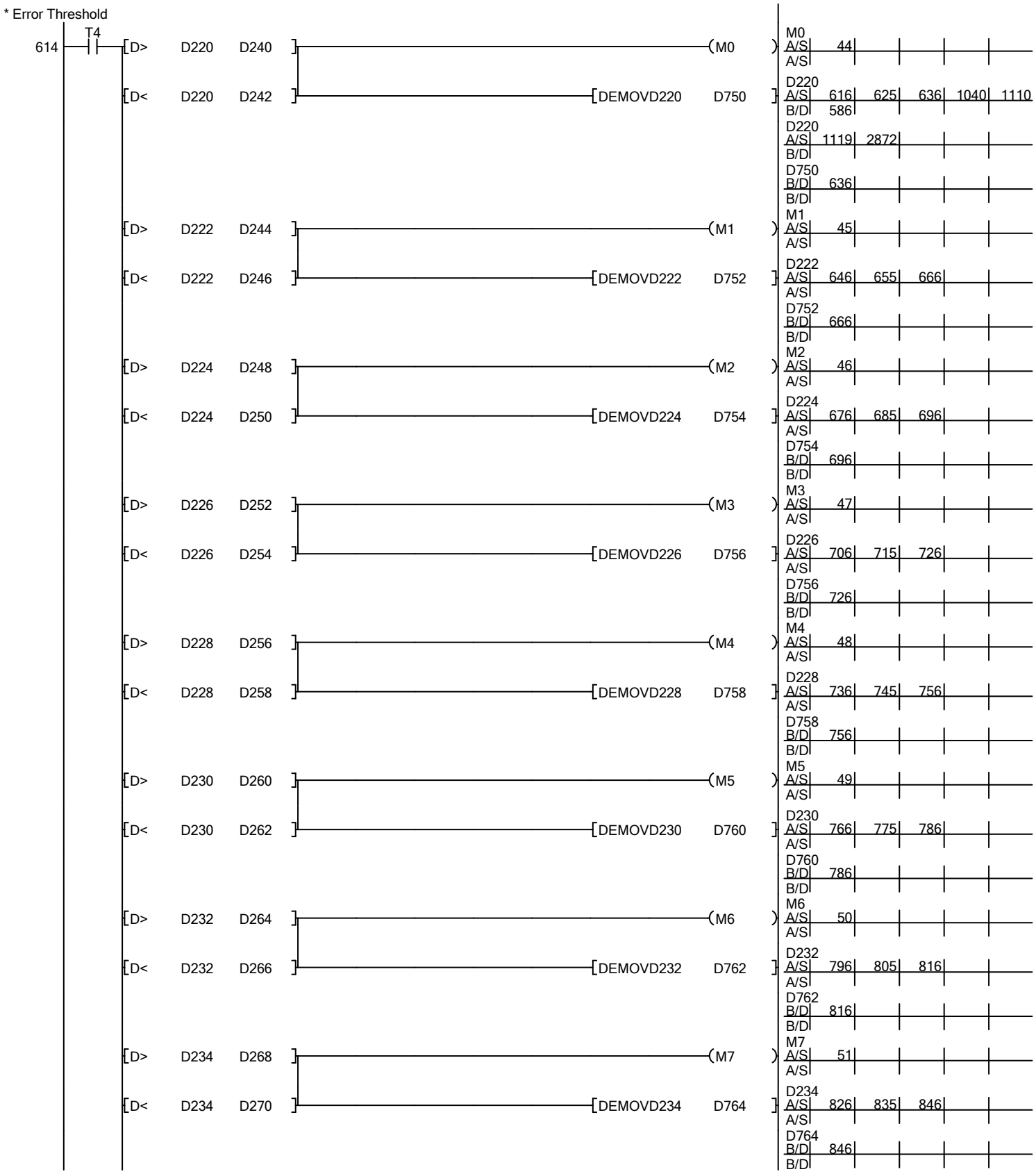

* ADC Float
167
M8000
FLT D0 D0
FLT D2 D2
FLT D102 D4
FLT D103 D6
FLT D104 D8
FLT D105 D10
FLT D106 D12
FLT D107 D14
FLT D108 D16
FLT D109 D18
FLT D20 D20
D0 A/S 168 225
B/D 134
D0 A/S 168 225
B/D 134
D2 A/S 173
B/D 139
D2 A/S 173
B/D 139
D102 A/S 178
B/D 149
D4 B/D 178
B/D
D103 A/S 183
A/S
D6 B/D 183
B/D
D104 A/S 188
A/S
D8 B/D 188
B/D
D105 A/S 193
A/S
D10 B/D 193
B/D
D106 A/S 198
B/D 158
D12 A/S 268
B/D 198
D107 A/S 203
A/S
D14 B/D 203
B/D
D108 A/S 208
A/S
D16 B/D 208
B/D
D109 A/S 213
A/S
D18 B/D 213
B/D
D20 A/S 218 310
B/D 144
D20 A/S 218 310
B/D 144

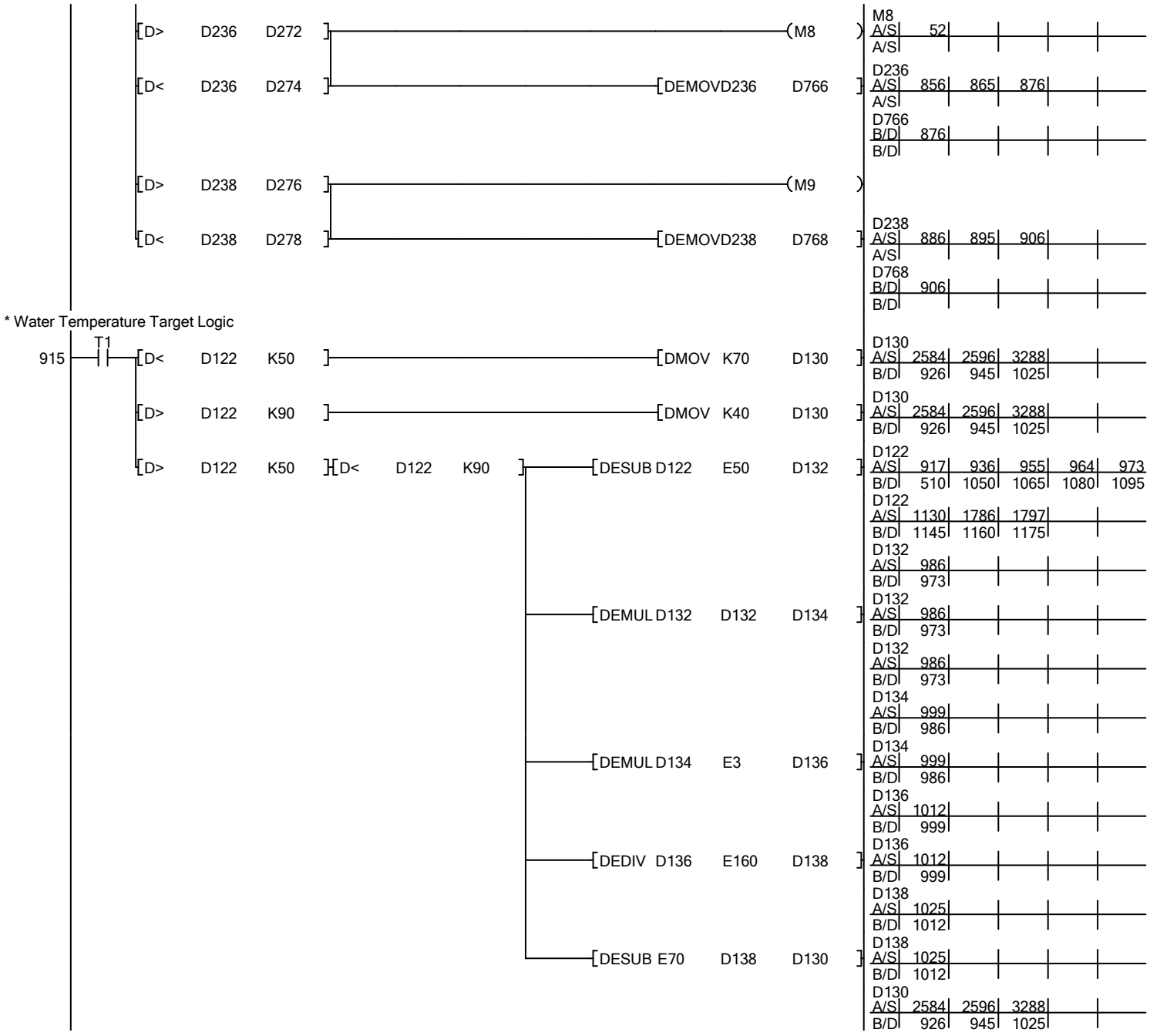

* [RTD to Temperature]
P0
T2
223
DEMUL D0Z0 E2.5 D100
DESUB D100 E1 D22Z0
ADD K2 Z0 Z0
258 < Z0 K12
CJ P0
* [Ultrasonic to Level]
P1
T2
266
DEMUL D12Z1 E2.5 D110
DESUB D110 E1 D34Z1
ADD K2 Z1 Z1
301 < Z1 K8
CJ P1
* Flowmeter to RPM
T0
309
DEMUL D20 E2.5 D112
DESUB D112 E1 D42
* [Sensor Averaging]
P2
T2
336
DEADD D22Z2 D44Z2 D44Z2
ADD K2 Z2 Z2
358 < Z2 K22
CJ P2

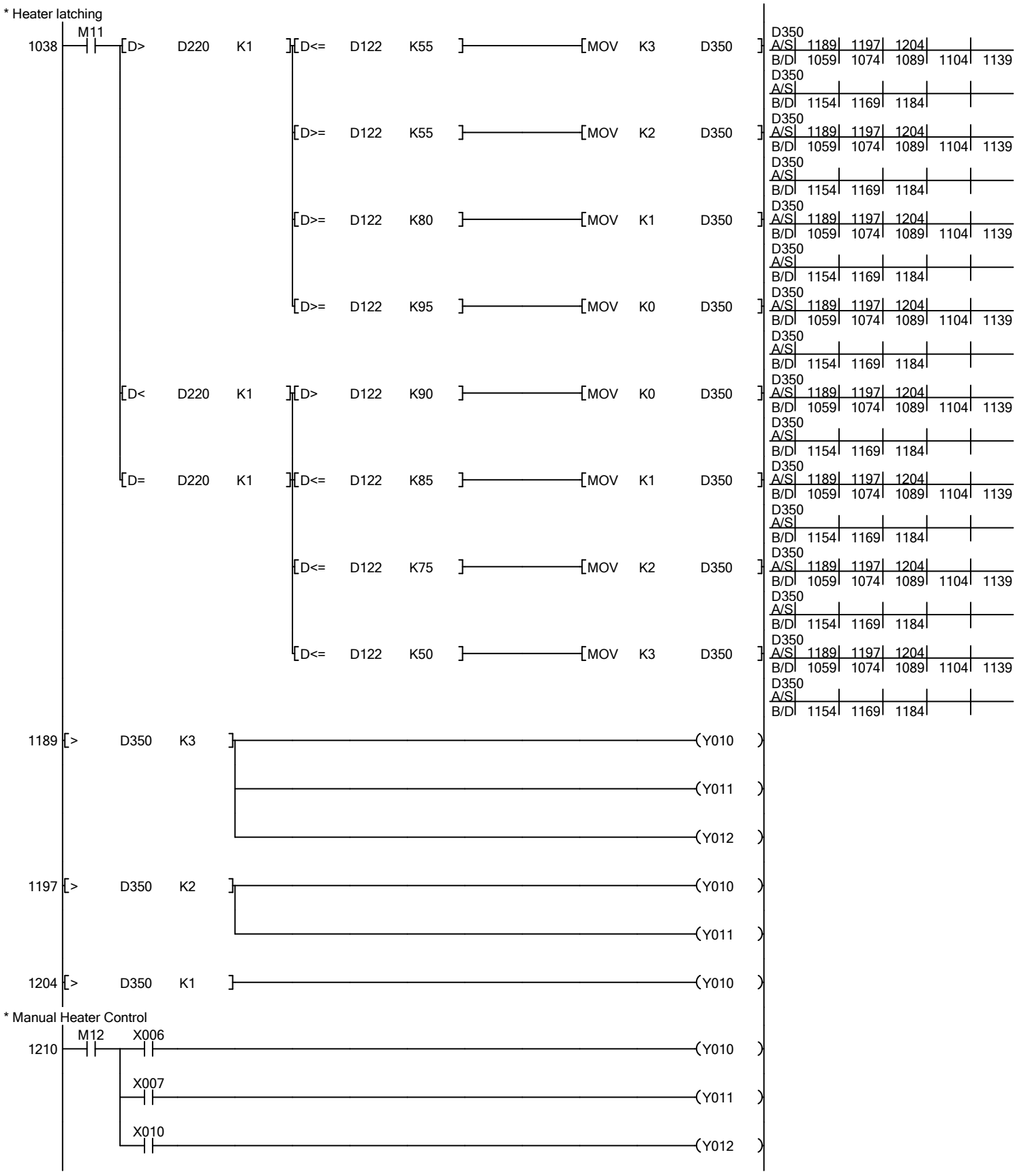

P3
366
T3
DEDIV D44Z3 E5 D64Z3
DEMOV E0 D44Z3
ADD K2 Z3 Z3
397
< Z3 K22
CJ P3
D44Z3
A/S 338 368
B/D 381
D64Z3
A/S 564 586 1274 1762 3283
B/D 368
D44Z3
A/S 338 368
B/D 381
Z3
A/S 368 390 397
B/D 101 381
Z3
A/S 368 390 397
B/D 101 381
P3
A/S 366 402
A/S

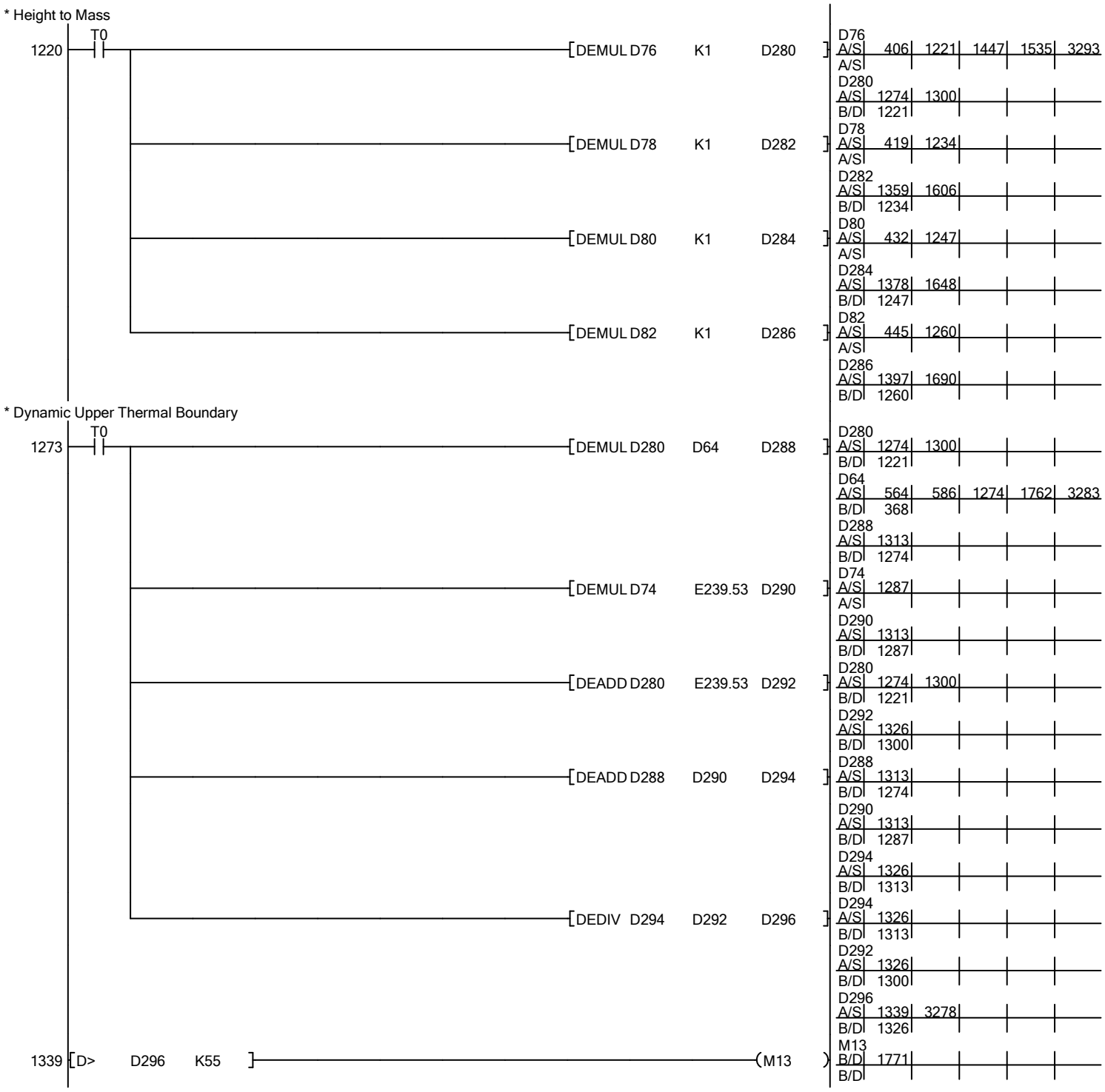

* Level % Converter
405
T1
DESUB K100 D76 D114
DESUB K100 D78 D116
DESUB K100 D80 D118
DESUB K100 D82 D120
DEDIV D114 K100 D114
DEDIV D116 K100 D116
DEDIV D118 K100 D118
DEDIV D120 K100 D120
DEMUL D114 K100 D122
DEMUL D116 K100 D124
DEMUL D118 K100 D126
DEMUL D120 K100 D128
D76 A/S 406 1221 1447 1535 3293
D114 A/S 458 510 B/D 406
D78 A/S 419 1234
D116 A/S 471 523 B/D 419
D80 A/S 432 1247
D118 A/S 484 536 B/D 432
D82 A/S 445 1260
D120 A/S 497 549 B/D 445
D114 A/S 458 510 B/D 406
D114 A/S 458 510 B/D 406
D116 A/S 471 523 B/D 419
D116 A/S 471 523 B/D 419
D118 A/S 484 536 B/D 432
D118 A/S 484 536 B/D 432
D120 A/S 497 549 B/D 445
D120 A/S 497 549 B/D 445
D114 A/S 458 510 B/D 406
D122 A/S 917 936 955 964 973 B/D 510 1050 1065 1080 1095
D122 A/S 1130 1786 1797 B/D 1145 1160 1175
D116 A/S 471 523 B/D 419
D124 B/D 523 B/D
D118 A/S 484 536 B/D 432
D126 B/D 536 B/D
D120 A/S 497 549 B/D 445
D128 B/D 549 B/D

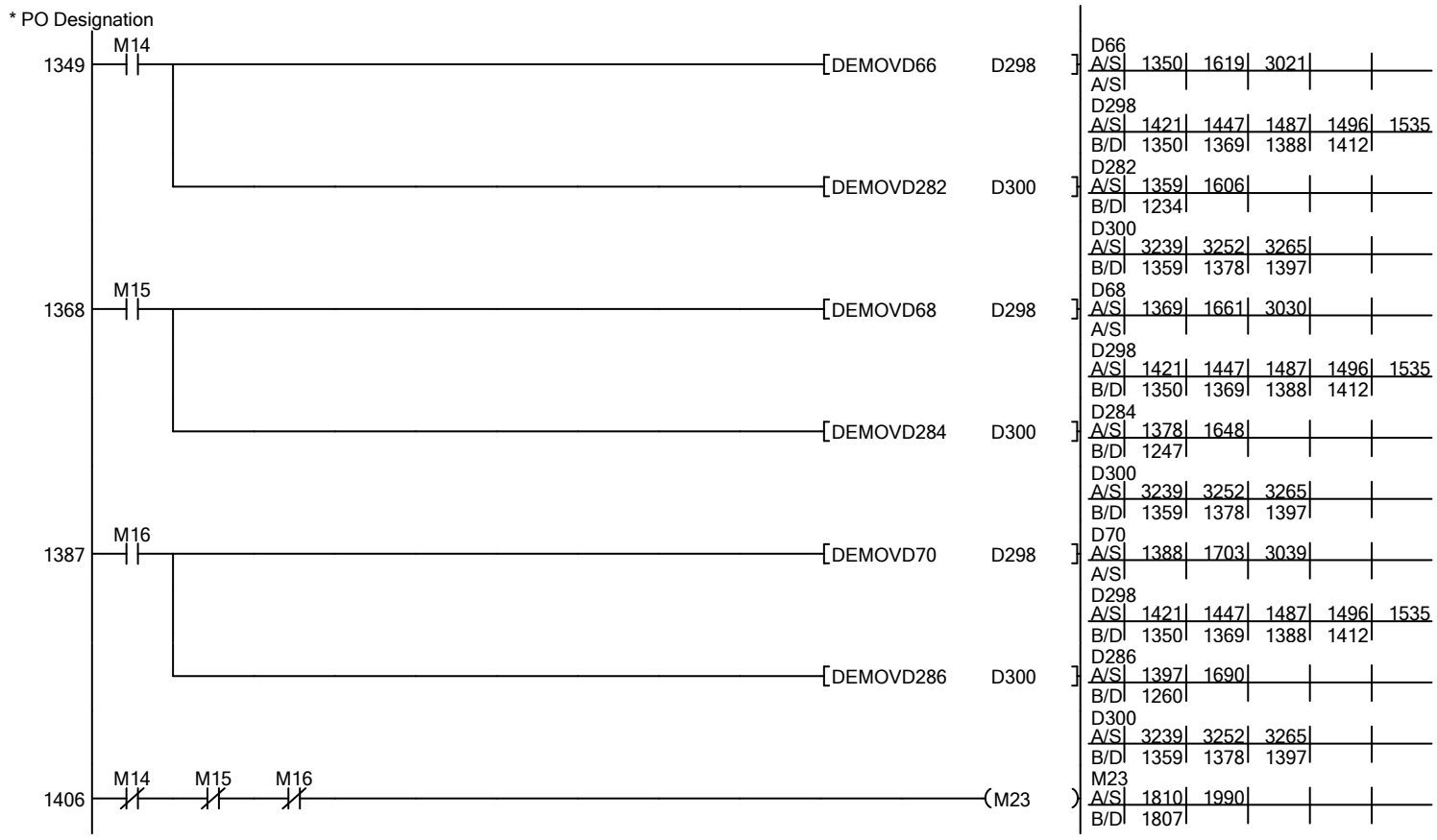

* [Sensor Logging (5 sec)]
P4
T3
562
MOV D64Z4 D200Z4
ADD K2 Z4 Z4
576 < Z4 K22
CJ P4
P5
T4
584
DEDIV D64Z5 D200Z5 D220Z5
ADD K2 Z5 Z5
606 < Z5 K20
CJ P5
D64Z4 A/S 564 586 1274 1762 3283 B/D 368
D200Z4 A/S 586 B/D 564
Z4 A/S 564 569 576 B/D 106
Z4 A/S 564 569 576 B/D 106
P4 A/S 562 581 A/S
D64Z5 A/S 564 586 1274 1762 3283 B/D 368
D200Z5 A/S 586 B/D 564
D220Z5 A/S 616 625 636 1040 1110 B/D 586
D220Z5 A/S 1119 2872 B/D
Z5 A/S 586 599 606 B/D 111
Z5 A/S 586 599 606 B/D 111
P5 A/S 584 611 A/S

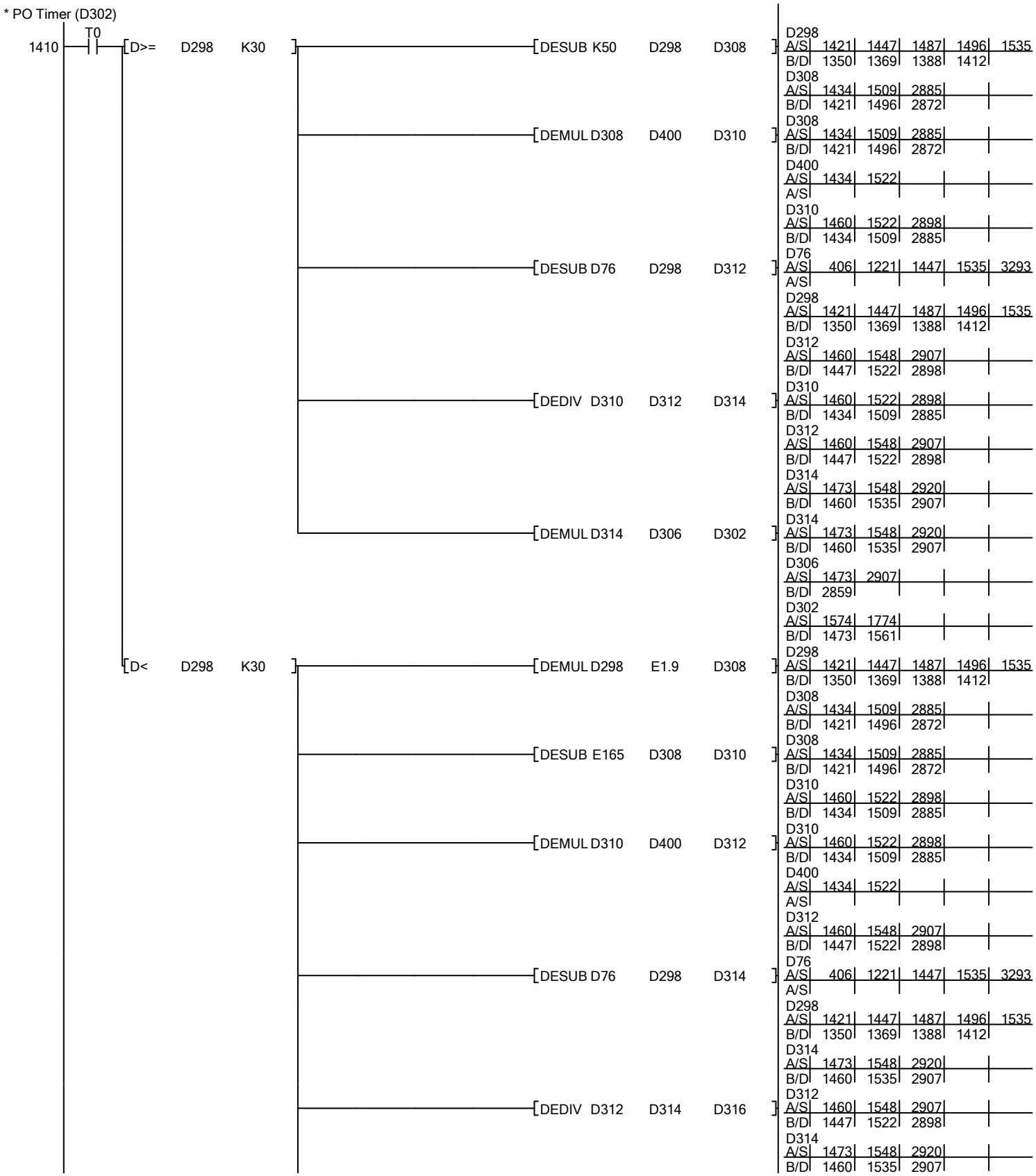

* Error Threshold
614
T4
D> D220 D240 M0
D< D220 D242 DEMOVD220 D750
D> D222 D244 M1
D< D222 D246 DEMOVD222 D752
D> D224 D248 M2
D< D224 D250 DEMOVD224 D754
D> D226 D252 M3
D< D226 D254 DEMOVD226 D756
D> D228 D256 M4
D< D228 D258 DEMOVD228 D758
D> D230 D260 M5
D< D230 D262 DEMOVD230 D760
D> D232 D264 M6
D< D232 D266 DEMOVD232 D762
D> D234 D268 M7
D< D234 D270 DEMOVD234 D764
M0 A/S 44 A/S
D220 A/S 616 625 636 1040 1110 B/D 586
D220 A/S 1119 2872 B/D
D750 B/D 636 B/D
M1 A/S 45 A/S
D222 A/S 646 655 666 A/S
D752 B/D 666 B/D
M2 A/S 46 A/S
D224 A/S 676 685 696 A/S
D754 B/D 696 B/D
M3 A/S 47 A/S
D226 A/S 706 715 726 A/S
D756 B/D 726 B/D
M4 A/S 48 A/S
D228 A/S 736 745 756 A/S
D758 B/D 756 B/D
M5 A/S 49 A/S
D230 A/S 766 775 786 A/S
D760 B/D 786 B/D
M6 A/S 50 A/S
D232 A/S 796 805 816 A/S
D762 B/D 816 B/D
M7 A/S 51 A/S
D234 A/S 826 835 846 A/S
D764 B/D 846 B/D

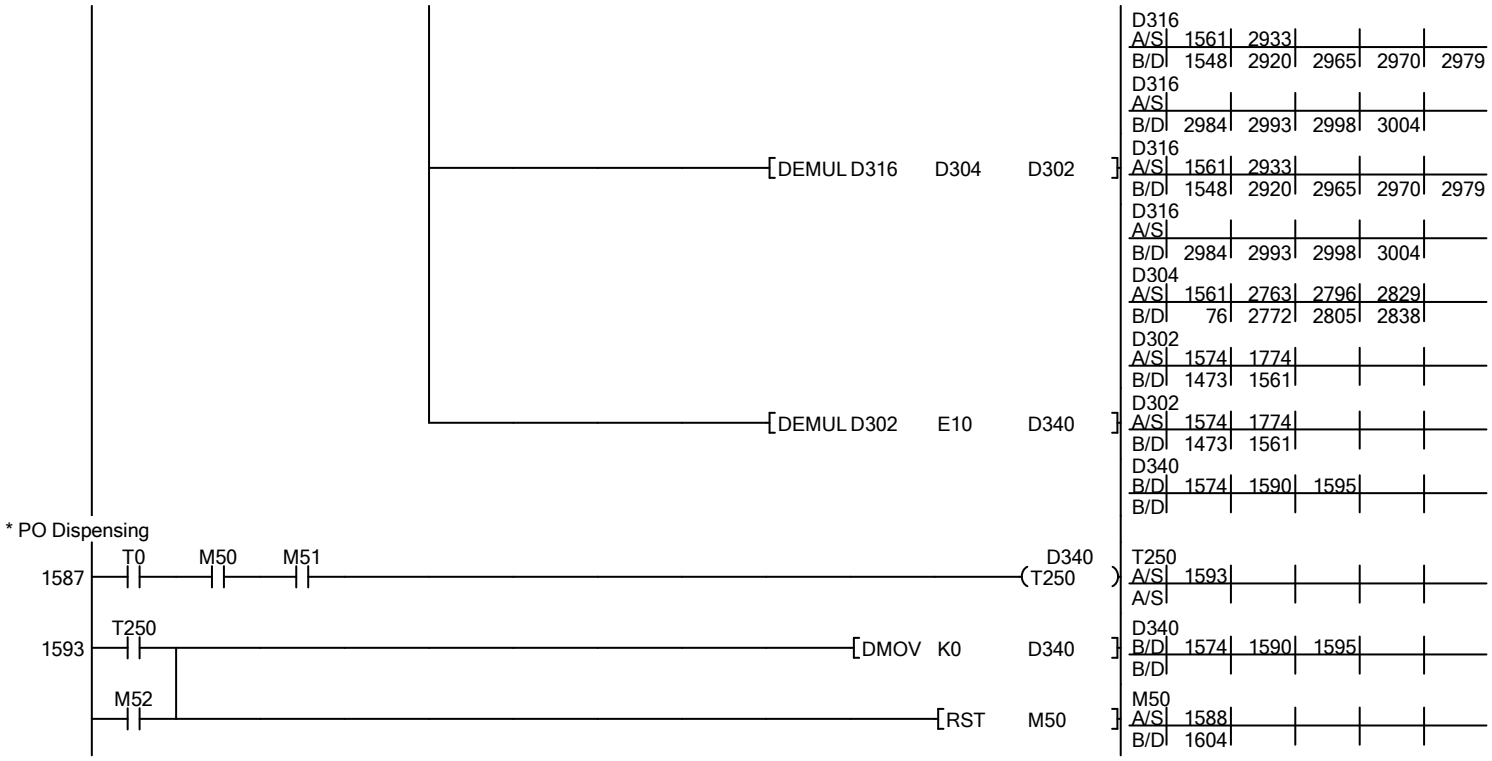

D> D236 D272
M8
D< D236 D274
DEMOV D236 D766
D> D238 D276
M9
D< D238 D278
DEMOV D238 D768
* Water Temperature Target Logic
915
T1
D< D122 K50
DMOV K70 D130
D> D122 K90
DMOV K40 D130
D> D122 K50
D< D122 K90
DESUB D122 E50 D132
DEMUL D132 D132 D134
DEMUL D134 E3 D136
DEDIV D136 E160 D138
DESUB E70 D138 D130

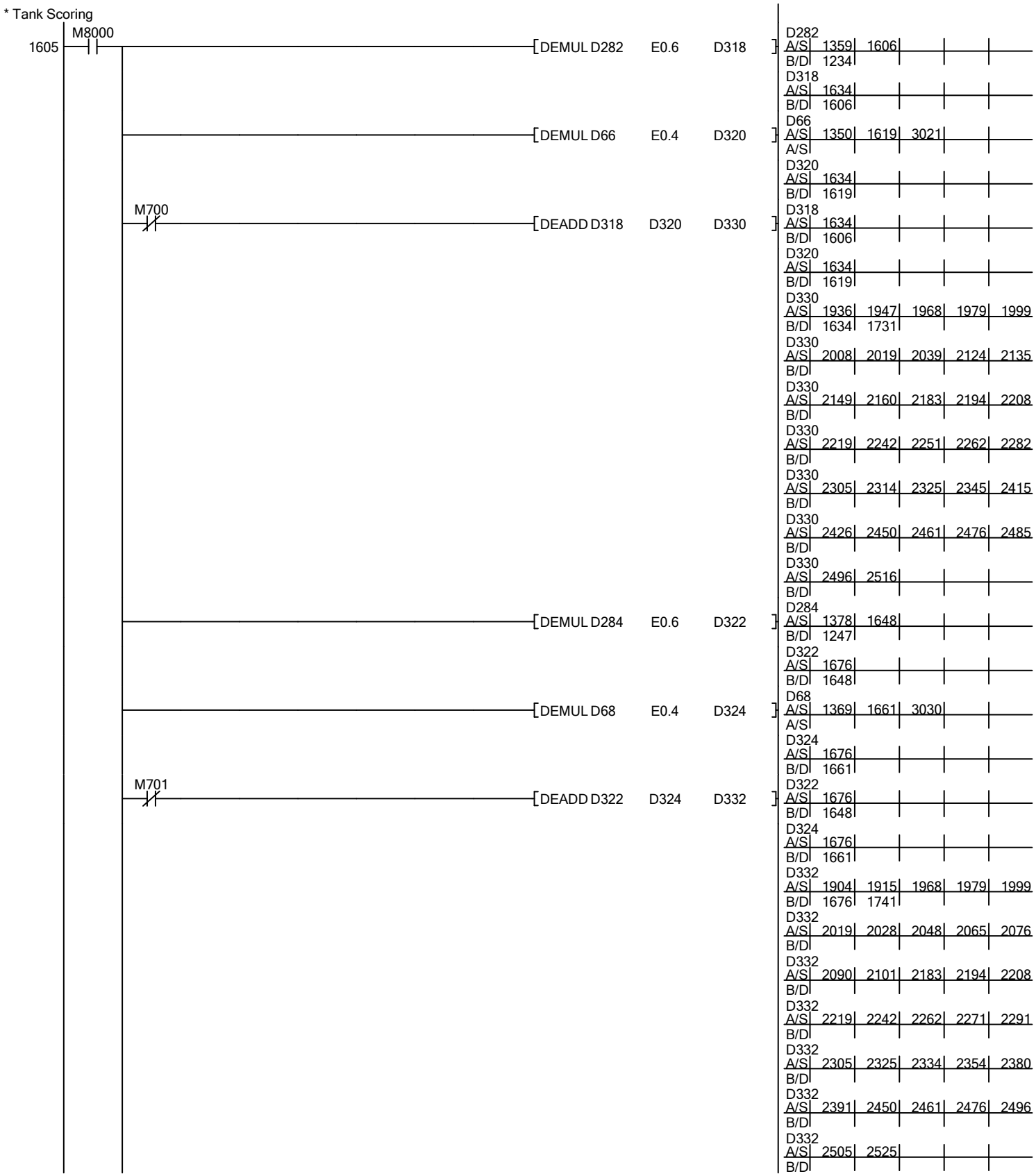

* Heater latching
1038 M11 D> D220 K1 D<= D122 K55 MOV K3 D350
D>= D122 K55 MOV K2 D350
D>= D122 K80 MOV K1 D350
D>= D122 K95 MOV K0 D350
D< D220 K1 D> D122 K90 MOV K0 D350
D= D220 K1 D<= D122 K85 MOV K1 D350
D<= D122 K75 MOV K2 D350
D<= D122 K50 MOV K3 D350
D350 A/S 1189 1197 1204 B/D 1059 1074 1089 1104 1139
D350 A/S B/D 1154 1169 1184
1189 > D350 K3 Y010 Y011 Y012
1197 > D350 K2 Y010 Y011
1204 > D350 K1 Y010
* Manual Heater Control
1210 M12 X006 Y010
X007 Y011
X010 Y012

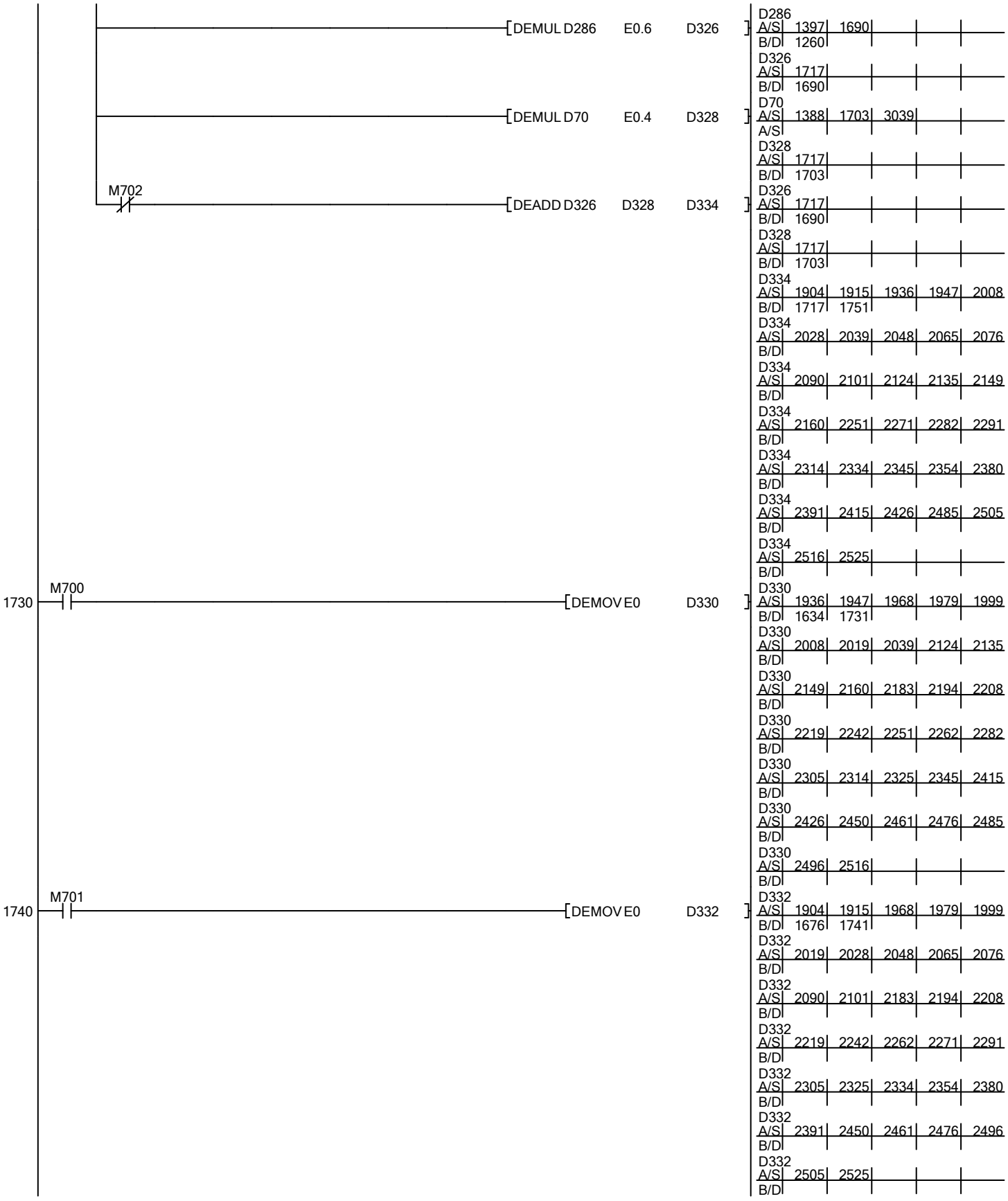

* Height to Mass
1220
T0
DEMUL D76 K1 D280
DEMUL D78 K1 D282
DEMUL D80 K1 D284
DEMUL D82 K1 D286
D76 A/S 406 1221 1447 1535 3293
D280 A/S 1274 1300 B/D 1221
D78 A/S 419 1234
D282 A/S 1359 1606 B/D 1234
D80 A/S 432 1247
D284 A/S 1378 1648 B/D 1247
D82 A/S 445 1260
D286 A/S 1397 1690 B/D 1260
* Dynamic Upper Thermal Boundary
1273
T0
DEMUL D280 D64 D288
DEMUL D74 E239.53 D290
DEADD D280 E239.53 D292
DEADD D288 D290 D294
DEDIV D294 D292 D296
D280 A/S 1274 1300 B/D 1221
D64 A/S 564 586 1274 1762 3283 B/D 368
D288 A/S 1313 B/D 1274
D74 A/S 1287
D290 A/S 1313 B/D 1287
D280 A/S 1274 1300 B/D 1221
D292 A/S 1326 B/D 1300
D288 A/S 1313 B/D 1274
D290 A/S 1313 B/D 1287
D294 A/S 1326 B/D 1313
D294 A/S 1326 B/D 1313
D292 A/S 1326 B/D 1300
D296 A/S 1339 3278 B/D 1326
1339 D> D296 K55
M13
M13 B/D 1771

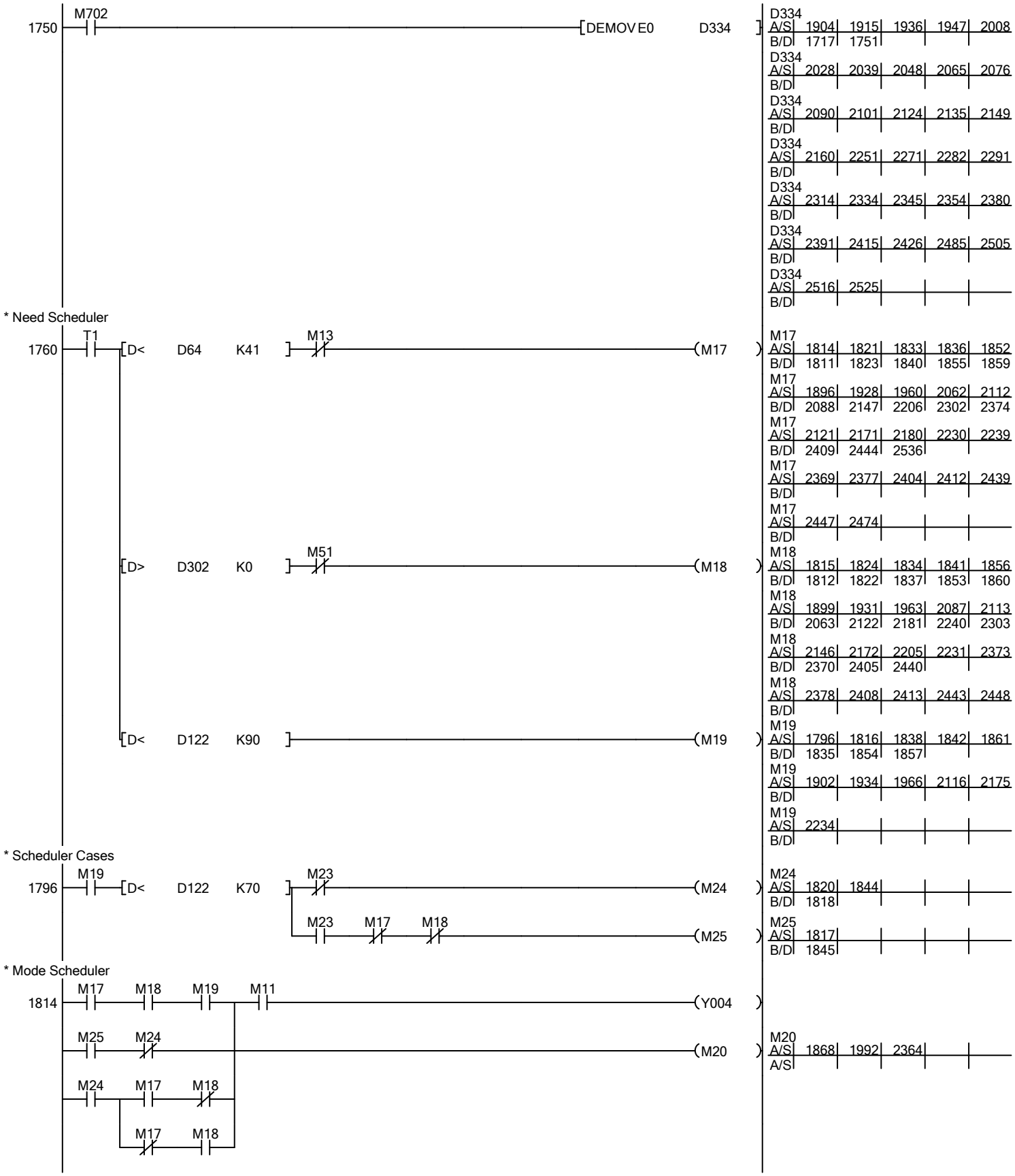

* PO Designation
M14
1349
DEMOVD66 D298
DEMOVD282 D300
M15
1368
DEMOVD68 D298
DEMOVD284 D300
M16
1387
DEMOVD70 D298
DEMOVD286 D300
M14 M15 M16
1406
M23
D66 A/S 1350 1619 3021 A/S
D298 A/S 1421 1447 1487 1496 1535 B/D 1350 1369 1388 1412
D282 A/S 1359 1606 B/D 1234
D300 A/S 3239 3252 3265 B/D 1359 1378 1397
D68 A/S 1369 1661 3030 A/S
D298 A/S 1421 1447 1487 1496 1535 B/D 1350 1369 1388 1412
D284 A/S 1378 1648 B/D 1247
D300 A/S 3239 3252 3265 B/D 1359 1378 1397
D70 A/S 1388 1703 3039 A/S
D298 A/S 1421 1447 1487 1496 1535 B/D 1350 1369 1388 1412
D286 A/S 1397 1690 B/D 1260
D300 A/S 3239 3252 3265 B/D 1359 1378 1397
M23 A/S 1810 1990 B/D 1807

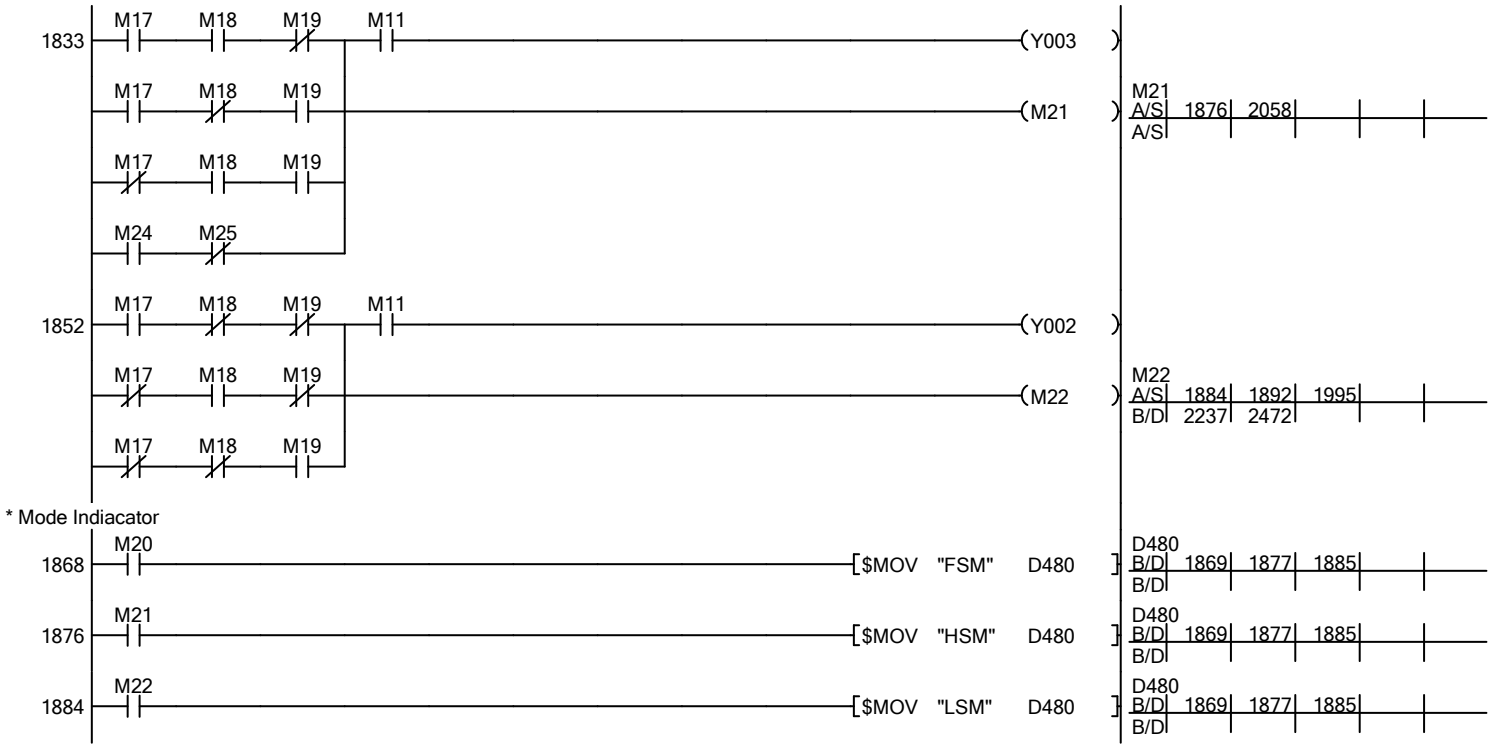

* PO Timer (D302)
1410
T0
D>= D298 K30
DESUB K50 D298 D308
DEMUL D308 D400 D310
DESUB D76 D298 D312
DEDIV D310 D312 D314
DEMUL D314 D306 D302
D< D298 K30
DEMUL D298 E1.9 D308
DESUB E165 D308 D310
DEMUL D310 D400 D312
DESUB D76 D298 D314
DEDIV D312 D314 D316

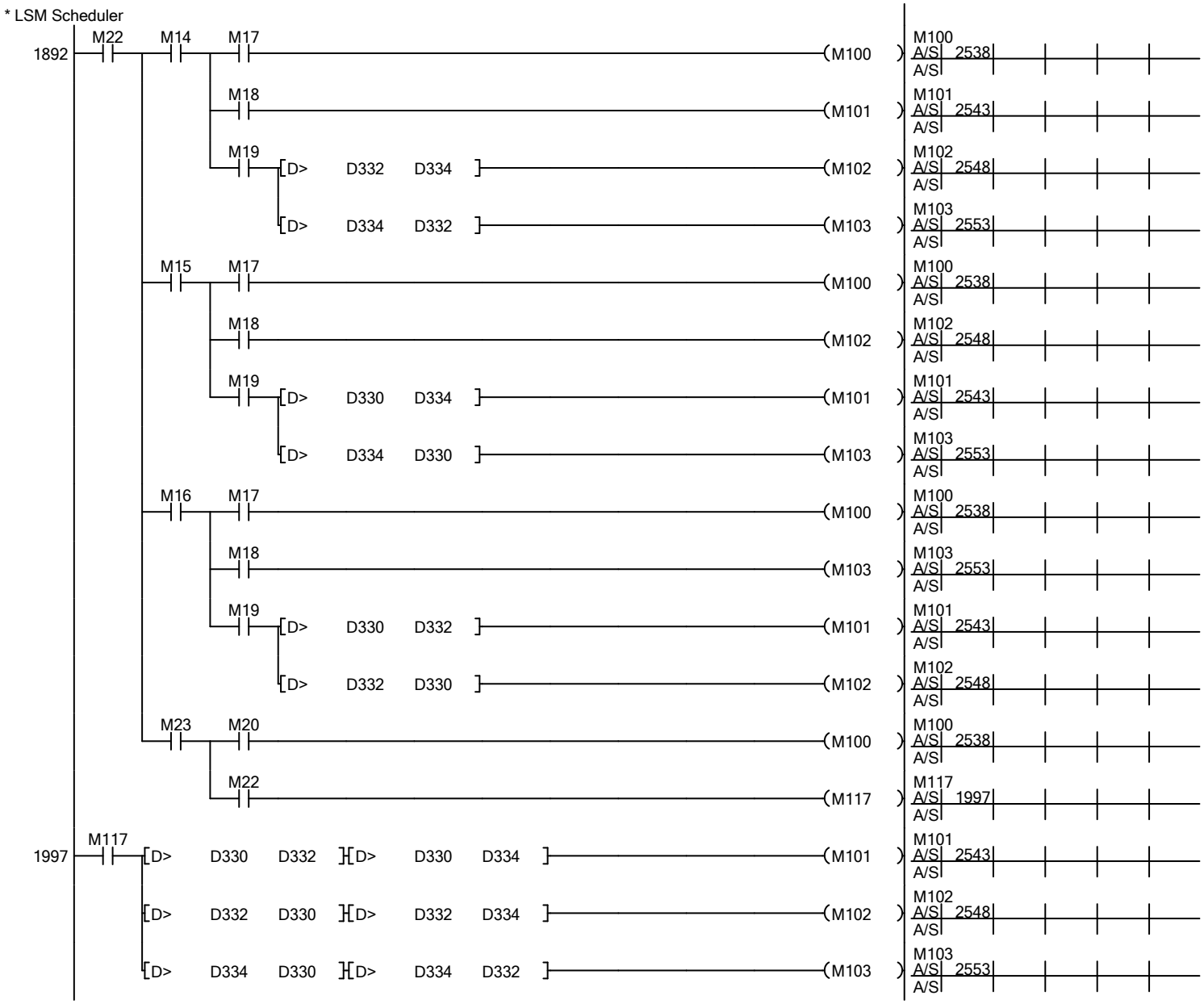

[DEMUL D316 D304 D302]
[DEMUL D302 E10 D340]
* PO Dispensing
1587
T0 M50 M51
D340
(T250)
1593
T250
[DMOV K0 D340]
M52
[RST M50]
D316 A/S 1561 2933 B/D 1548 2920 2965 2970 2979
D316 A/S B/D 2984 2993 2998 3004
D316 A/S 1561 2933 B/D 1548 2920 2965 2970 2979
D316 A/S B/D 2984 2993 2998 3004
D304 A/S 1561 2763 2796 2829 B/D 76 2772 2805 2838
D302 A/S 1574 1774 B/D 1473 1561
D302 A/S 1574 1774 B/D 1473 1561
D340 B/D 1574 1590 1595 B/D
T250 A/S 1593 A/S
D340 B/D 1574 1590 1595 B/D
M50 A/S 1588 B/D 1604

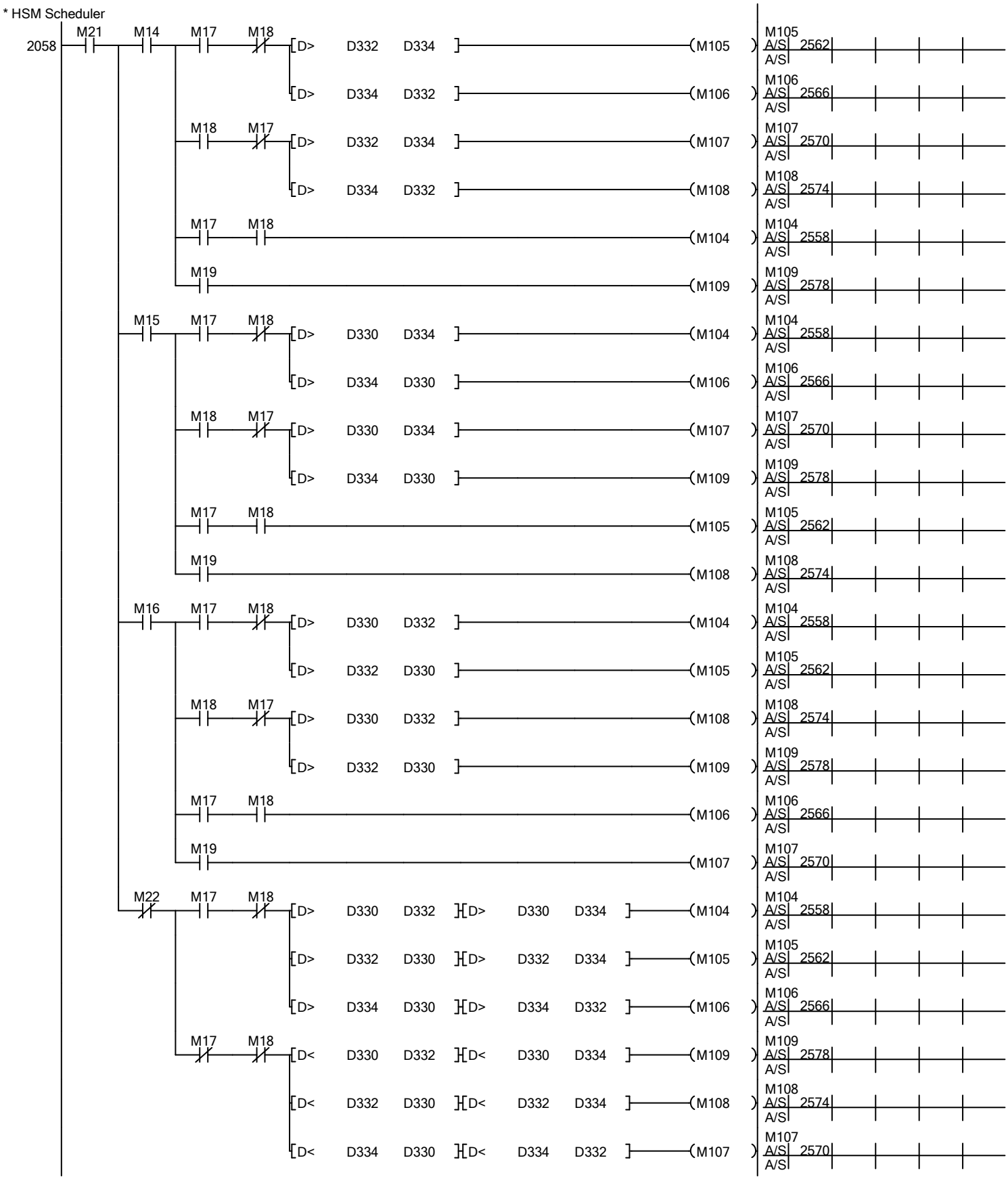

* Tank Scoring
1605
M8000
DEMUL D282 E0.6 D318
D282 A/S 1359 1606 B/D 1234
D318 A/S 1634 B/D 1606
DEMUL D66 E0.4 D320
D66 A/S 1350 1619 3021 A/S
D320 A/S 1634 B/D 1619
M700
DEADD D318 D320 D330
D318 A/S 1634 B/D 1606
D320 A/S 1634 B/D 1619
D330 A/S 1936 1947 1968 1979 1999 B/D 1634 1731
D330 A/S 2008 2019 2039 2124 2135 B/D
D330 A/S 2149 2160 2183 2194 2208 B/D
D330 A/S 2219 2242 2251 2262 2282 B/D
D330 A/S 2305 2314 2325 2345 2415 B/D
D330 A/S 2426 2450 2461 2476 2485 B/D
D330 A/S 2496 2516 B/D
DEMUL D284 E0.6 D322
D284 A/S 1378 1648 B/D 1247
D322 A/S 1676 B/D 1648
DEMUL D68 E0.4 D324
D68 A/S 1369 1661 3030 A/S
D324 A/S 1676 B/D 1661
M701
DEADD D322 D324 D332
D322 A/S 1676 B/D 1648
D324 A/S 1676 B/D 1661
D332 A/S 1904 1915 1968 1979 1999 B/D 1676 1741
D332 A/S 2019 2028 2048 2065 2076 B/D
D332 A/S 2090 2101 2183 2194 2208 B/D
D332 A/S 2219 2242 2262 2271 2291 B/D
D332 A/S 2305 2325 2334 2354 2380 B/D
D332 A/S 2391 2450 2461 2476 2496 B/D
D332 A/S 2505 2525 B/D

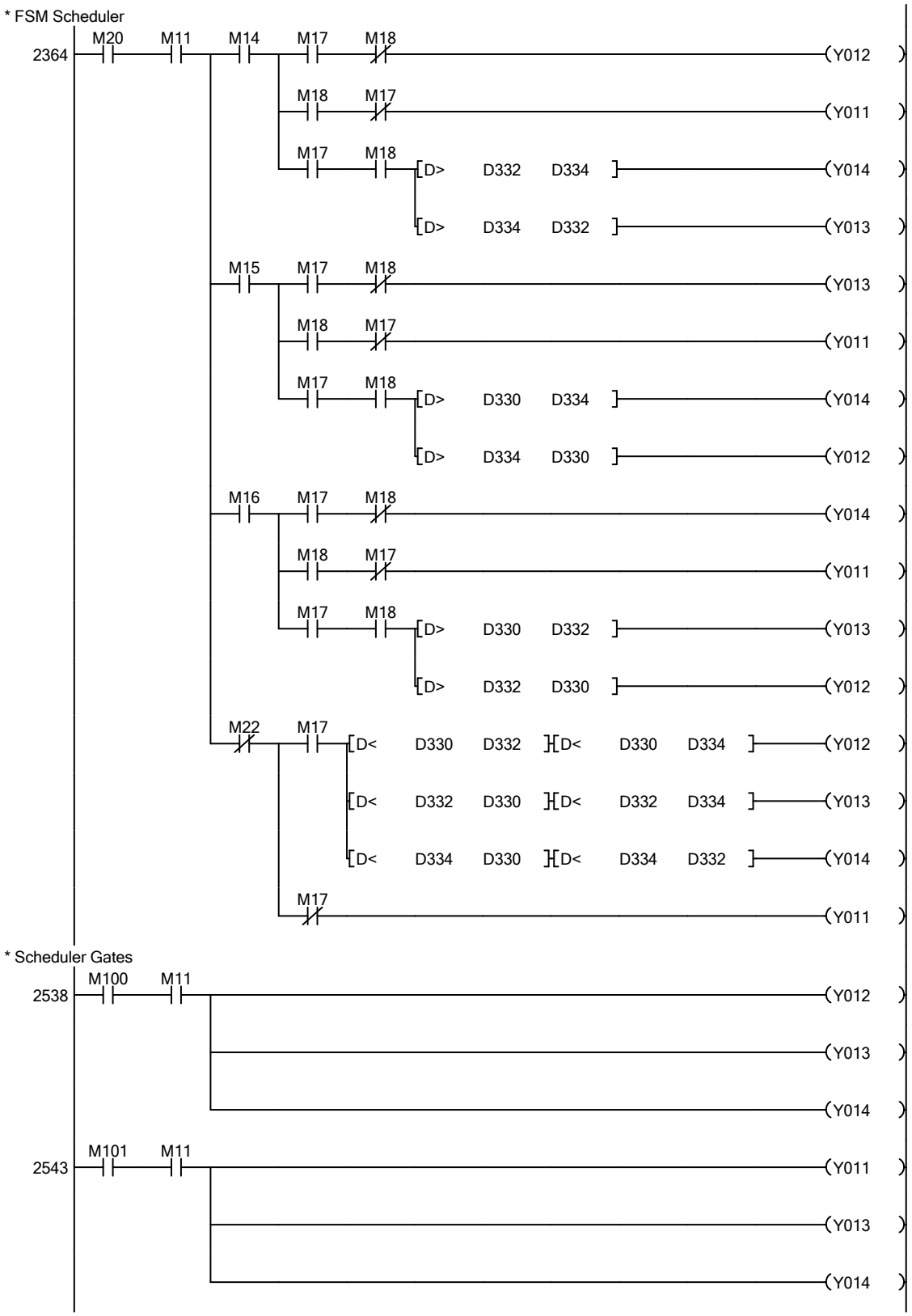

DEMUL D286 E0.6 D326
D286
A/S 1397 1690
B/D 1260
D326
A/S 1717
B/D 1690
DEMUL D70 E0.4 D328
D70
A/S 1388 1703 3039
A/S
D328
A/S 1717
B/D 1703
M702
DEADD D326 D328 D334
D326
A/S 1717
B/D 1690
D328
A/S 1717
B/D 1703
D334
A/S 1904 1915 1936 1947 2008
B/D 1717 1751
D334
A/S 2028 2039 2048 2065 2076
B/D
D334
A/S 2090 2101 2124 2135 2149
B/D
D334
A/S 2160 2251 2271 2282 2291
B/D
D334
A/S 2314 2334 2345 2354 2380
B/D
D334
A/S 2391 2415 2426 2485 2505
B/D
D334
A/S 2516 2525
B/D
M700
1730
DEMOV E0 D330
D330
A/S 1936 1947 1968 1979 1999
B/D 1634 1731
D330
A/S 2008 2019 2039 2124 2135
B/D
D330
A/S 2149 2160 2183 2194 2208
B/D
D330
A/S 2219 2242 2251 2262 2282
B/D
D330
A/S 2305 2314 2325 2345 2415
B/D
D330
A/S 2426 2450 2461 2476 2485
B/D
D330
A/S 2496 2516
B/D
M701
1740
DEMOV E0 D332
D332
A/S 1904 1915 1968 1979 1999
B/D 1676 1741
D332
A/S 2019 2028 2048 2065 2076
B/D
D332
A/S 2090 2101 2183 2194 2208
B/D
D332
A/S 2219 2242 2262 2271 2291
B/D
D332
A/S 2305 2325 2334 2354 2380
B/D
D332
A/S 2391 2450 2461 2476 2496
B/D
D332
A/S 2505 2525
B/D

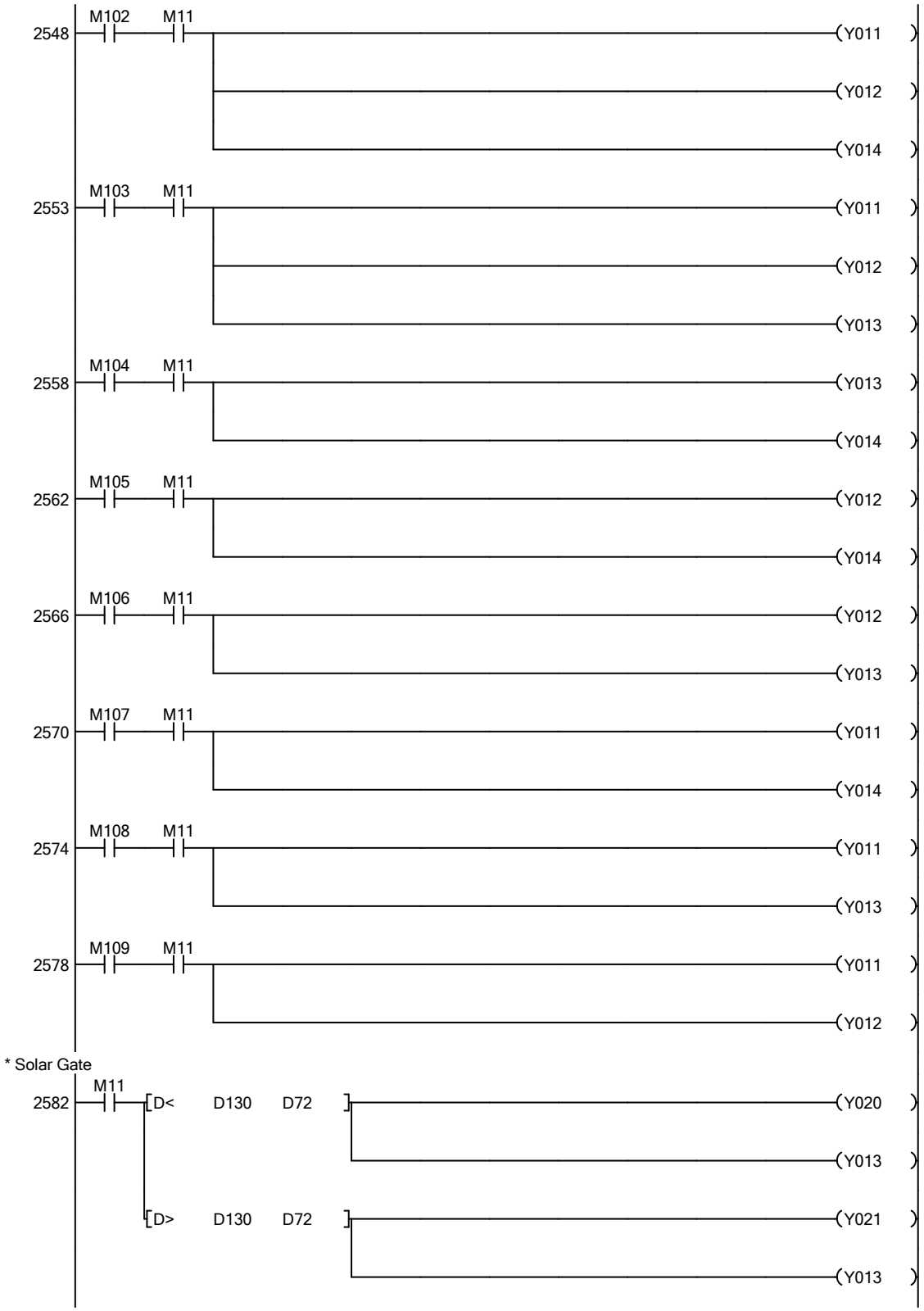

1750
M702
DEMOVE0
D334
D334 A/S 1904 1915 1936 1947 2008 B/D 1717 1751
D334 A/S 2028 2039 2048 2065 2076 B/D
D334 A/S 2090 2101 2124 2135 2149 B/D
D334 A/S 2160 2251 2271 2282 2291 B/D
D334 A/S 2314 2334 2345 2354 2380 B/D
D334 A/S 2391 2415 2426 2485 2505 B/D
D334 A/S 2516 2525 B/D
* Need Scheduler
1760
T1
D< D64 K41
M13
M17
M17 A/S 1814 1821 1833 1836 1852 B/D 1811 1823 1840 1855 1859
M17 A/S 1896 1928 1960 2062 2112 B/D 2088 2147 2206 2302 2374
M17 A/S 2121 2171 2180 2230 2239 B/D 2409 2444 2536
M17 A/S 2369 2377 2404 2412 2439 B/D
M17 A/S 2447 2474 B/D
D> D302 K0
M51
M18
M18 A/S 1815 1824 1834 1841 1856 B/D 1812 1822 1837 1853 1860
M18 A/S 1899 1931 1963 2087 2113 B/D 2063 2122 2181 2240 2303
M18 A/S 2146 2172 2205 2231 2373 B/D 2370 2405 2440
M18 A/S 2378 2408 2413 2443 2448 B/D
D< D122 K90
M19
M19 A/S 1796 1816 1838 1842 1861 B/D 1835 1854 1857
M19 A/S 1902 1934 1966 2116 2175 B/D
M19 A/S 2234 B/D
* Scheduler Cases
1796
M19
D< D122 K70
M23
M24
M24 A/S 1820 1844 B/D 1818
M23 M17 M18
M25
M25 A/S 1817 B/D 1845
* Mode Scheduler
1814
M17 M18 M19 M11
Y004
M25 M24
M20
M20 A/S 1868 1992 2364 A/S
M24 M17 M18
M17 M18

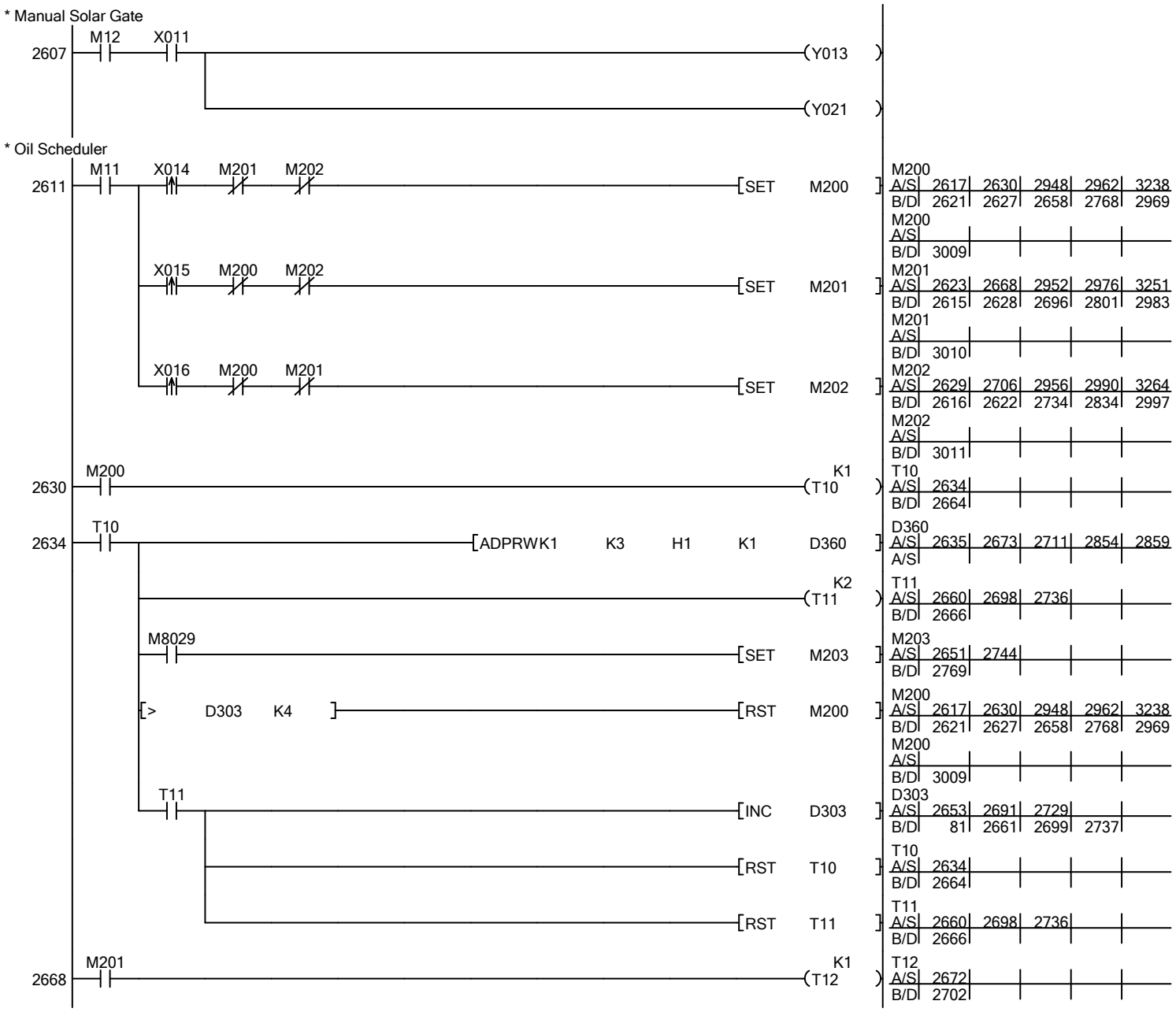

1833
M17 M18 M19 M11
Y003
M17 M18 M19
M21
M21
A/S 1876 2058
A/S
M17 M18 M19
M24 M25
1852
M17 M18 M19 M11
Y002
M17 M18 M19
M22
M22
A/S 1884 1892 1995
B/D 2237 2472
M17 M18 M19
* Mode Indiacator
1868
M20
[$MOV "FSM" D480]
D480
B/D 1869 1877 1885
B/D
1876
M21
[$MOV "HSM" D480]
D480
B/D 1869 1877 1885
B/D
1884
M22
[$MOV "LSM" D480]
D480
B/D 1869 1877 1885
B/D

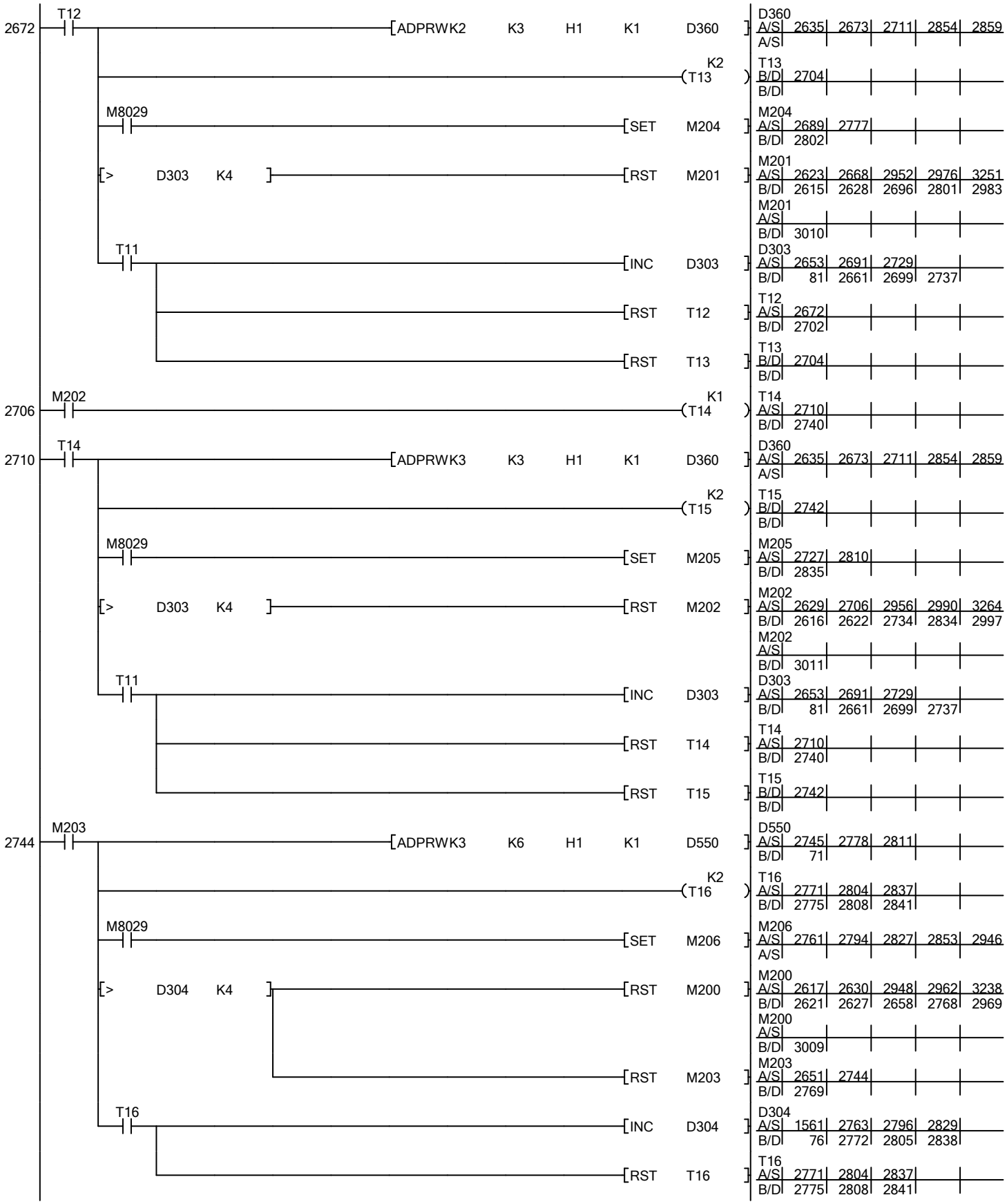

* LSM Scheduler
1892
M22
M14
M17
M100
M100 A/S 2538 A/S
M18
M101
M101 A/S 2543 A/S
M19
D> D332 D334
M102
M102 A/S 2548 A/S
D> D334 D332
M103
M103 A/S 2553 A/S
M15
M17
M100
M100 A/S 2538 A/S
M18
M102
M102 A/S 2548 A/S
M19
D> D330 D334
M101
M101 A/S 2543 A/S
D> D334 D330
M103
M103 A/S 2553 A/S
M16
M17
M100
M100 A/S 2538 A/S
M18
M103
M103 A/S 2553 A/S
M19
D> D330 D332
M101
M101 A/S 2543 A/S
D> D332 D330
M102
M102 A/S 2548 A/S
M23
M20
M100
M100 A/S 2538 A/S
M22
M117
M117 A/S 1997 A/S
1997
M117
D> D330 D332 D> D330 D334
M101
M101 A/S 2543 A/S
D> D332 D330 D> D332 D334
M102
M102 A/S 2548 A/S
D> D334 D330 D> D334 D332
M103
M103 A/S 2553 A/S

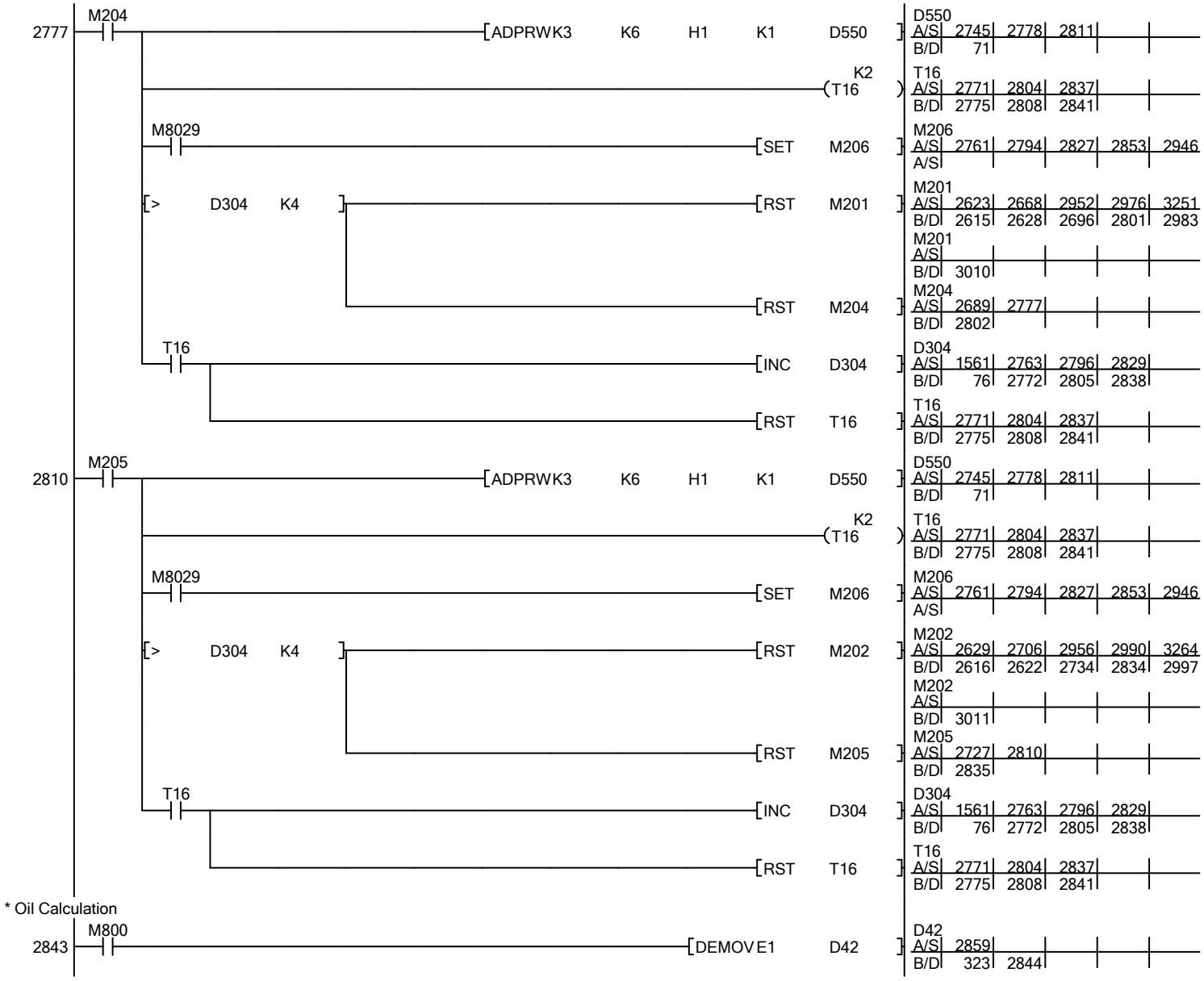

* HSM Scheduler
2058
M21 M14 M17 M18 [D> D332 D334] (M105)
[D> D334 D332] (M106)
M18 M17 [D> D332 D334] (M107)
[D> D334 D332] (M108)
M17 M18 (M104)
M19 (M109)
M15 M17 M18 [D> D330 D334] (M104)
[D> D334 D330] (M106)
M18 M17 [D> D330 D334] (M107)
[D> D334 D330] (M109)
M17 M18 (M105)
M19 (M108)
M16 M17 M18 [D> D330 D332] (M104)
[D> D332 D330] (M105)
M18 M17 [D> D330 D332] (M108)
[D> D332 D330] (M109)
M17 M18 (M106)
M19 (M107)
M22 M17 M18 [D> D330 D332][D> D330 D334] (M104)
[D> D332 D330][D> D332 D334] (M105)
[D> D334 D330][D> D334 D332] (M106)
M17 M18 [D< D330 D332][D< D330 D334] (M109)
[D< D332 D330][D< D332 D334] (M108)
[D< D334 D330][D< D334 D332] (M107)
M105 A/S 2562 A/S
M106 A/S 2566 A/S
M107 A/S 2570 A/S
M108 A/S 2574 A/S
M104 A/S 2558 A/S
M109 A/S 2578 A/S

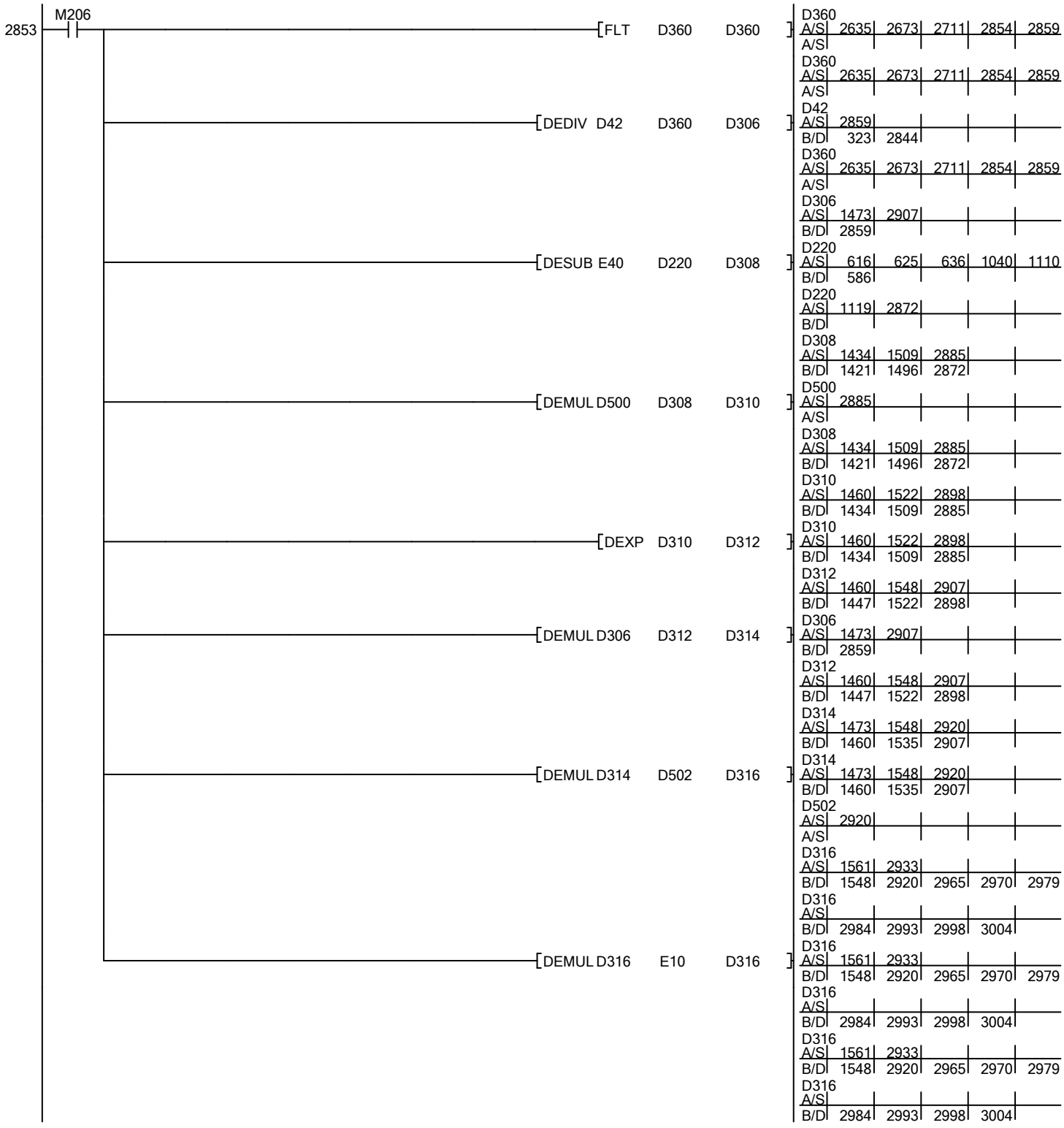

* FSM Scheduler
2364
M20 M11 M14 M17 M18 (Y012)
M18 M17 (Y011)
M17 M18 [D> D332 D334] (Y014)
[D> D334 D332] (Y013)
M15 M17 M18 (Y013)
M18 M17 (Y011)
M17 M18 [D> D330 D334] (Y014)
[D> D334 D330] (Y012)
M16 M17 M18 (Y014)
M18 M17 (Y011)
M17 M18 [D> D330 D332] (Y013)
[D> D332 D330] (Y012)
M22 M17 [D< D330 D332][D< D330 D334] (Y012)
[D< D332 D330][D< D332 D334] (Y013)
[D< D334 D330][D< D334 D332] (Y014)
M17 (Y011)
* Scheduler Gates
2538
M100 M11 (Y012)
(Y013)
(Y014)
2543
M101 M11 (Y011)
(Y013)
(Y014)

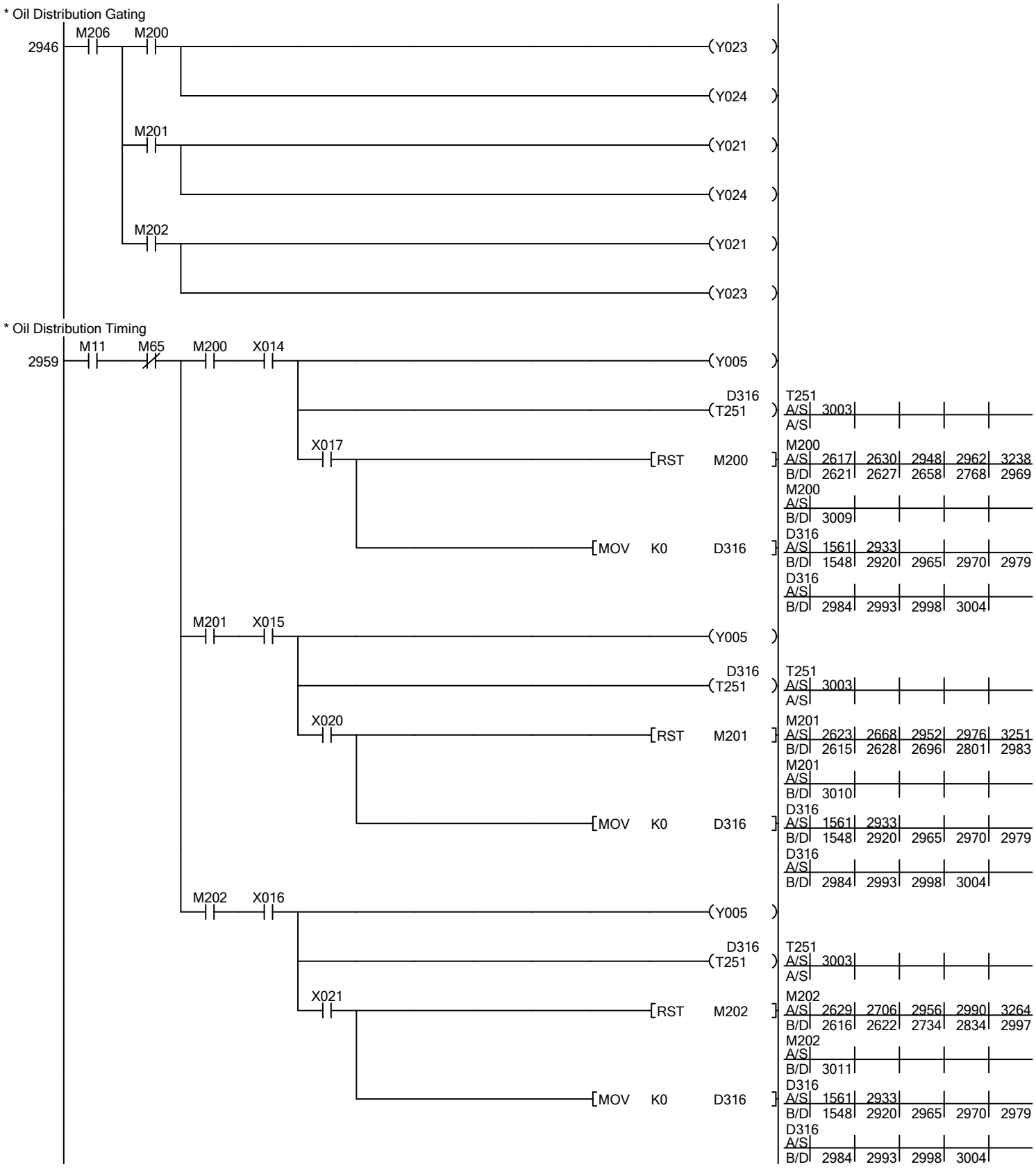

2548
M102
M11
Y011
Y012
Y014
2553
M103
M11
Y011
Y012
Y013
2558
M104
M11
Y013
Y014
2562
M105
M11
Y012
Y014
2566
M106
M11
Y012
Y013
2570
M107
M11
Y011
Y014
2574
M108
M11
Y011
Y013
2578
M109
M11
Y011
Y012
* Solar Gate
2582
M11
D< D130 D72
Y020
Y013
D> D130 D72
Y021
Y013

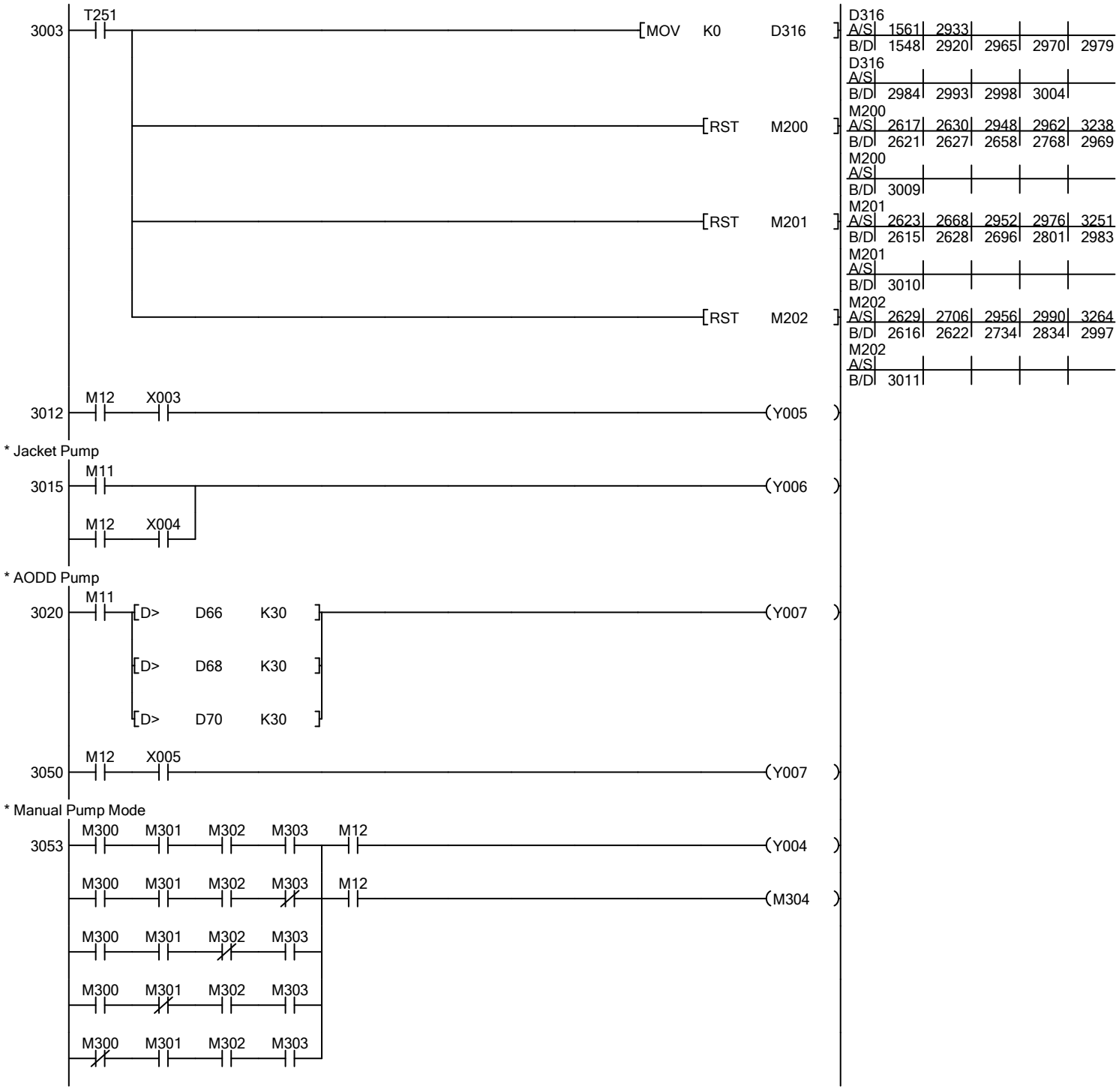

* Manual Solar Gate
2607
M12
X011
Y013
Y021
* Oil Scheduler
2611
M11
X014
M201
M202
SET M200
X015
M200
M202
SET M201
X016
M200
M201
SET M202
2630
M200
K1
T10
2634
T10
ADPRWK1 K3 H1 K1 D360
K2
T11
M8029
SET M203
> D303 K4
RST M200
T11
INC D303
RST T10
RST T11
2668
M201
K1
T12
M200 A/S 2617 2630 2948 2962 3238 B/D 2621 2627 2658 2768 2969
M200 A/S B/D 3009
M201 A/S 2623 2668 2952 2976 3251 B/D 2615 2628 2696 2801 2983
M201 A/S B/D 3010
M202 A/S 2629 2706 2956 2990 3264 B/D 2616 2622 2734 2834 2997
M202 A/S B/D 3011
T10 A/S 2634 B/D 2664
D360 A/S 2635 2673 2711 2854 2859 A/S
T11 A/S 2660 2698 2736 B/D 2666
M203 A/S 2651 2744 B/D 2769
M200 A/S 2617 2630 2948 2962 3238 B/D 2621 2627 2658 2768 2969
M200 A/S B/D 3009
D303 A/S 2653 2691 2729 B/D 81 2661 2699 2737
T10 A/S 2634 B/D 2664
T11 A/S 2660 2698 2736 B/D 2666
T12 A/S 2672 B/D 2702

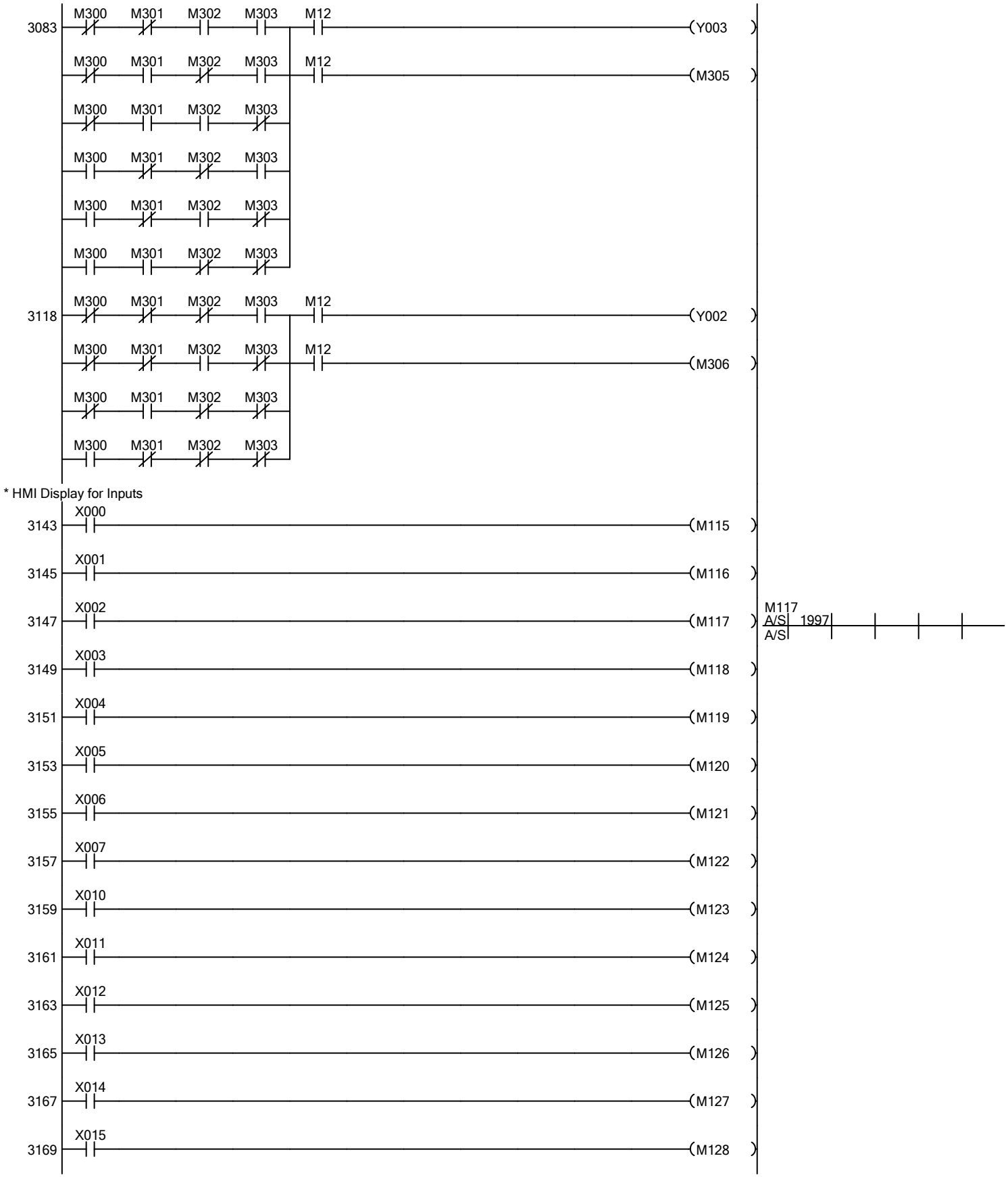

2672
T12
ADPRW K2 K3 H1 K1 D360
K2 T13
M8029
SET M204
> D303 K4
RST M201
T11
INC D303
RST T12
RST T13
M202
2706
K1 T14
T14
2710
ADPRW K3 K3 H1 K1 D360
K2 T15
M8029
SET M205
> D303 K4
RST M202
T11
INC D303
RST T14
RST T15
M203
2744
ADPRW K3 K6 H1 K1 D550
K2 T16
M8029
SET M206
> D304 K4
RST M200
RST M203
T16
INC D304
RST T16
D360 A/S 2635 2673 2711 2854 2859
T13 B/D 2704
M204 A/S 2689 2777 B/D 2802
M201 A/S 2623 2668 2952 2976 3251 B/D 2615 2628 2696 2801 2983
M201 B/D 3010
D303 A/S 2653 2691 2729 B/D 81 2661 2699 2737
T12 A/S 2672 B/D 2702
T13 B/D 2704
T14 A/S 2710 B/D 2740
D360 A/S 2635 2673 2711 2854 2859
T15 B/D 2742
M205 A/S 2727 2810 B/D 2835
M202 A/S 2629 2706 2956 2990 3264 B/D 2616 2622 2734 2834 2997
M202 B/D 3011
D303 A/S 2653 2691 2729 B/D 81 2661 2699 2737
T14 A/S 2710 B/D 2740
T15 B/D 2742
D550 A/S 2745 2778 2811 B/D 71
T16 A/S 2771 2804 2837 B/D 2775 2808 2841
M206 A/S 2761 2794 2827 2853 2946
M200 A/S 2617 2630 2948 2962 3238 B/D 2621 2627 2658 2768 2969
M200 A/S B/D 3009
M203 A/S 2651 2744 B/D 2769
D304 A/S 1561 2763 2796 2829 B/D 76 2772 2805 2838
T16 A/S 2771 2804 2837 B/D 2775 2808 2841

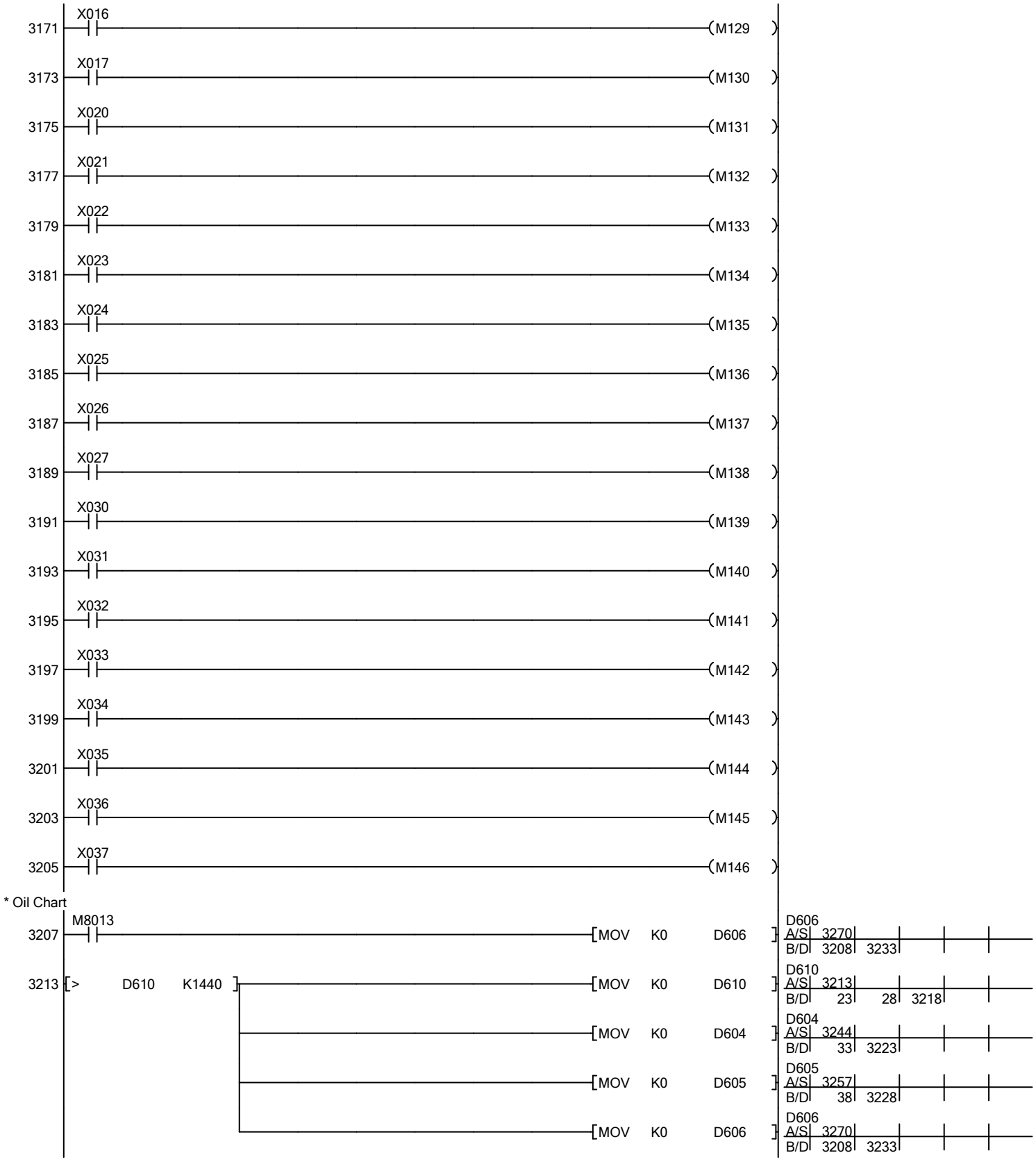

2777
M204
ADPRW K3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M201
RST M204
T16
INC D304
RST T16
2810
M205
ADPRW K3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M202
RST M205
T16
INC D304
RST T16
* Oil Calculation
2843
M800
DEMOV E1 D42
D550 A/S 2745 2778 2811 B/D 71
T16 A/S 2771 2804 2837 B/D 2775 2808 2841
M206 A/S 2761 2794 2827 2853 2946 A/S
M201 A/S 2623 2668 2952 2976 3251 B/D 2615 2628 2696 2801 2983
M201 A/S B/D 3010
M204 A/S 2689 2777 B/D 2802
D304 A/S 1561 2763 2796 2829 B/D 76 2772 2805 2838
T16 A/S 2771 2804 2837 B/D 2775 2808 2841
D550 A/S 2745 2778 2811 B/D 71
T16 A/S 2771 2804 2837 B/D 2775 2808 2841
M206 A/S 2761 2794 2827 2853 2946 A/S
M202 A/S 2629 2706 2956 2990 3264 B/D 2616 2622 2734 2834 2997
M202 A/S B/D 3011
M205 A/S 2727 2810 B/D 2835
D304 A/S 1561 2763 2796 2829 B/D 76 2772 2805 2838
T16 A/S 2771 2804 2837 B/D 2775 2808 2841
D42 A/S 2859 B/D 323 2844

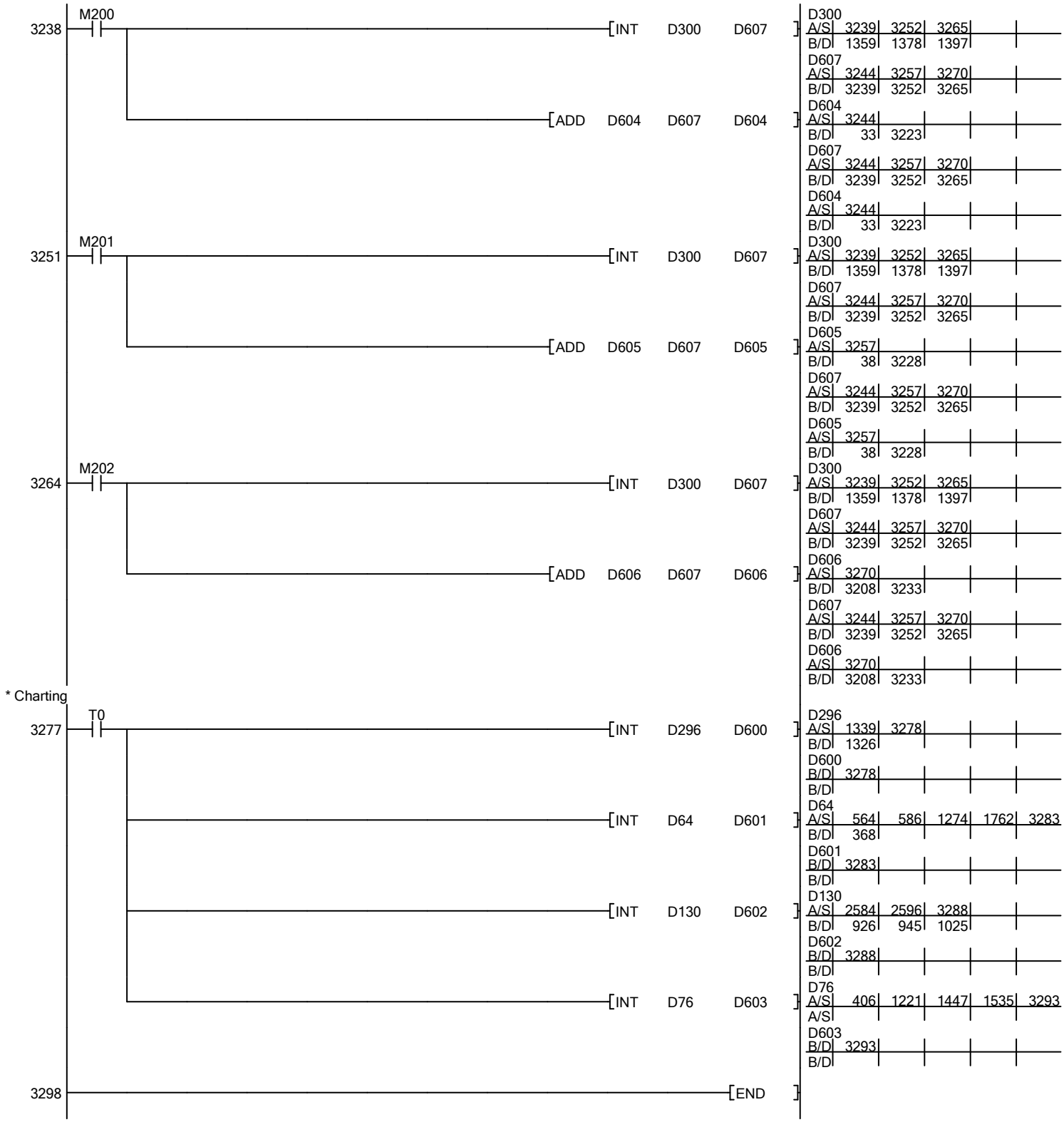

2853
M206
FLT D360 D360
D360 A/S 2635 2673 2711 2854 2859 A/S
D360 A/S 2635 2673 2711 2854 2859 A/S
DEDIV D42 D360 D306
D42 A/S 2859 B/D 323 2844
D360 A/S 2635 2673 2711 2854 2859 A/S
D306 A/S 1473 2907 B/D 2859
DESUB E40 D220 D308
D220 A/S 616 625 636 1040 1110 B/D 586
D220 A/S 1119 2872 B/D
D308 A/S 1434 1509 2885 B/D 1421 1496 2872
DEMUL D500 D308 D310
D500 A/S 2885 A/S
D308 A/S 1434 1509 2885 B/D 1421 1496 2872
D310 A/S 1460 1522 2898 B/D 1434 1509 2885
DEXP D310 D312
D310 A/S 1460 1522 2898 B/D 1434 1509 2885
D312 A/S 1460 1548 2907 B/D 1447 1522 2898
DEMUL D306 D312 D314
D306 A/S 1473 2907 B/D 2859
D312 A/S 1460 1548 2907 B/D 1447 1522 2898
D314 A/S 1473 1548 2920 B/D 1460 1535 2907
DEMUL D314 D502 D316
D314 A/S 1473 1548 2920 B/D 1460 1535 2907
D502 A/S 2920 A/S
D316 A/S 1561 2933 B/D 1548 2920 2965 2970 2979
D316 A/S B/D 2984 2993 2998 3004
DEMUL D316 E10 D316
D316 A/S 1561 2933 B/D 1548 2920 2965 2970 2979
D316 A/S B/D 2984 2993 2998 3004
D316 A/S 1561 2933 B/D 1548 2920 2965 2970 2979
D316 A/S B/D 2984 2993 2998 3004


* Oil Distribution Gating
2946
M206
M200
Y023
Y024
M201
Y021
Y024
M202
Y021
Y023
* Oil Distribution Timing
2959
M11
M65
M200
X014
Y005
D316
T251
T251
A/S 3003
A/S
X017
RST M200
M200
A/S 2617 2630 2948 2962 3238
B/D 2621 2627 2658 2768 2969
M200
A/S
B/D 3009
MOV K0 D316
D316
A/S 1561 2933
B/D 1548 2920 2965 2970 2979
D316
A/S
B/D 2984 2993 2998 3004
M201
X015
Y005
D316
T251
T251
A/S 3003
A/S
X020
RST M201
M201
A/S 2623 2668 2952 2976 3251
B/D 2615 2628 2696 2801 2983
M201
A/S
B/D 3010
MOV K0 D316
D316
A/S 1561 2933
B/D 1548 2920 2965 2970 2979
D316
A/S
B/D 2984 2993 2998 3004
M202
X016
Y005
D316
T251
T251
A/S 3003
A/S
X021
RST M202
M202
A/S 2629 2706 2956 2990 3264
B/D 2616 2622 2734 2834 2997
M202
A/S
B/D 3011
MOV K0 D316
D316
A/S 1561 2933
B/D 1548 2920 2965 2970 2979
D316
A/S
B/D 2984 2993 2998 3004


3003
T251
MOV K0 D316
RST M200
RST M201
RST M202
D316
A/S 1561 2933
B/D 1548 2920 2965 2970 2979
D316
A/S
B/D 2984 2993 2998 3004
M200
A/S 2617 2630 2948 2962 3238
B/D 2621 2627 2658 2768 2969
M200
A/S
B/D 3009
M201
A/S 2623 2668 2952 2976 3251
B/D 2615 2628 2696 2801 2983
M201
A/S
B/D 3010
M202
A/S 2629 2706 2956 2990 3264
B/D 2616 2622 2734 2834 2997
M202
A/S
B/D 3011
3012
M12 X003
Y005
* Jacket Pump
3015
M11
M12 X004
Y006
* AODD Pump
3020
M11
D> D66 K30
D> D68 K30
D> D70 K30
Y007
3050
M12 X005
Y007
* Manual Pump Mode
3053
M300 M301 M302 M303 M12
Y004
M300 M301 M302 M303 M12
M304
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303


3083
M300 M301 M302 M303 M12 Y003
M300 M301 M302 M303 M12 M305
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
3118
M300 M301 M302 M303 M12 Y002
M300 M301 M302 M303 M12 M306
M300 M301 M302 M303
M300 M301 M302 M303
* HMI Display for Inputs
3143 X000 M115
3145 X001 M116
3147 X002 M117
M117 A/S 1997 A/S
3149 X003 M118
3151 X004 M119
3153 X005 M120
3155 X006 M121
3157 X007 M122
3159 X010 M123
3161 X011 M124
3163 X012 M125
3165 X013 M126
3167 X014 M127
3169 X015 M128


3171 X016 M129
3173 X017 M130
3175 X020 M131
3177 X021 M132
3179 X022 M133
3181 X023 M134
3183 X024 M135
3185 X025 M136
3187 X026 M137
3189 X027 M138
3191 X030 M139
3193 X031 M140
3195 X032 M141
3197 X033 M142
3199 X034 M143
3201 X035 M144
3203 X036 M145
3205 X037 M146
* Oil Chart
3207 M8013 [MOV K0 D606]
D606 A/S 3270 B/D 3208 3233
3213 [> D610 K1440]
[MOV K0 D610]
D610 A/S 3213 B/D 23 28 3218
[MOV K0 D604]
D604 A/S 3244 B/D 33 3223
[MOV K0 D605]
D605 A/S 3257 B/D 38 3228
[MOV K0 D606]
D606 A/S 3270 B/D 3208 3233


3238
M200
INT D300 D607
ADD D604 D607 D604
3251
M201
INT D300 D607
ADD D605 D607 D605
3264
M202
INT D300 D607
ADD D606 D607 D606
* Charting
3277
T0
INT D296 D600
INT D64 D601
INT D130 D602
INT D76 D603
3298
END
D300 A/S 3239 3252 3265 B/D 1359 1378 1397
D607 A/S 3244 3257 3270 B/D 3239 3252 3265
D604 A/S 3244 B/D 33 3223
D605 A/S 3257 B/D 38 3228
D606 A/S 3270 B/D 3208 3233
D296 A/S 1339 3278 B/D 1326
D600 B/D 3278
D64 A/S 564 586 1274 1762 3283 B/D 368
D601 B/D 3283
D130 A/S 2584 2596 3288 B/D 926 945 1025
D602 B/D 3288
D76 A/S 406 1221 1447 1535 3293
D603 B/D 3293

[Line Statement List]

Step	Line Statement
0	Timers
22	Load Once
43	E-Stops
56	Automation
65	Declarations
133	Analog to Digital
167	ADC Float
309	Flowmeter to RPM
405	Level % Converter
614	Error Threshold
915	Water Temperature Target Logic
1038	Heater latching
1210	Manual Heater Control
1220	Height to Mass
1273	Dynamic Upper Thermal Boundary
1349	PO Designation
1410	PO Timer (D302)
1587	PO Dispensing
1605	Tank Scoring
1760	Need Scheduler
1796	Scheduler Cases
1814	Mode Scheduler
1868	Mode Indiacator
1892	LSM Scheduler
2058	HSM Scheduler
2364	FSM Scheduler
2538	Scheduler Gates
2582	Solar Gate
2607	Manual Solar Gate
2611	Oil Scheduler
2843	Oil Calculation
2946	Oil Distribution Gating
2959	Oil Distribution Timing
3015	Jacket Pump
3020	AODD Pump
3053	Manual Pump Mode
3143	HMI Display for Inputs
3207	Oil Chart
3277	Charting

[P Statement List]

Device	P Statement
P0	RTD to Temperature
P1	Ultrasonic to Level
P2	Sensor Averaging
P4	Sensor Logging (5 sec)